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The RHOMOLO interim macroeconomic analysis of the 2021-2027 European Social Fund Plus

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Abstract

We estimate the macroeconomic impact of the 2021-2027 European Social Fund Plus (ESF+) investments that fall under the broad areas of employment, education and training, and social inclusion with data updated to the end of 2024. We employ the spatial dynamic general equilibrium model RHOMOLO, which has been extended to incorporate endogenous labour force participation. The investments serving as inputs to the model affect the labour supply or labour productivity in each Member State. The results suggest that ESF+ funds are expected to lead to positive GDP, labour force participation and employment responses across all Member States, while cross-country disparities decline.

Keywords: European Social Fund, labour markets, general equilibrium modelling.

JEL Codes: C68, J20, J30, R13.

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Executive summary

The European Social Fund Plus (ESF+) is a key instrument of the European Union's cohesion policy. It aims to promote high employment levels, fair social protection and a skilled, resilient workforce. This report presents an interim macroeconomic impact assessment of the 2021–27 ESF+ programme, which was conducted using a spatial dynamic general equilibrium model. The assessment focuses on three broad areas of intervention: employment; education and training; and social inclusion.

Key Findings

The results suggest that the ESF+ will have a positive impact on the GDP of Member States, with an estimated increase of 0.17% by the time policy implementation ends in 2029, remaining substantial also in the long run. However, the impact varies across countries, with the largest positive changes expected in Eastern and Southern European Member States. ESF+ interventions are expected to generate a positive return on investment, with a GDP multiplier of €2.28 for every €1 invested at the EU level by 2040. The policy is also expected to have a positive impact on labour force entry and employment, with an additional 411,000 individuals entering the labour force and 303,000 new workers in 2040. Furthermore, the ESF+ is expected to reduce disparities both between and within countries, evidenced by a decline in the coefficient of variation and the interpercentile ratio of GDP per capita.

Policy Implications

The findings of this report have significant implications for policymakers and stakeholders involved in implementing and evaluating the ESF+. The positive impact of the ESF+ on GDP, labour market entry and employment suggests that the policy is effective in achieving its objectives. The heterogeneous impact across countries highlights the need for tailored interventions that consider the specific needs and characteristics of each Member State. The reduction in disparities across the EU indicates that the ESF+ could contribute to a more cohesive and inclusive labour market. Overall, the report provides policymakers with valuable insights to help them refine the implementation of the ESF+ and maximise its impact on the labour market and the economy.

1. Introduction

The European Social Fund Plus (ESF+) is a key instrument of the European Union's cohesion policy for the 2021-2027 programming period. Building on the successes of its predecessors, the ESF+ aims to promote high employment levels, fair social protection and a skilled and resilient workforce ready for the future world of work, as well as inclusive and cohesive societies aiming to eradicating poverty (Regulation (EU) 1057/2021). With a total budget of over €142.7 billion, the ESF+ supports a range of investments in human capital, social inclusion, and labour market initiatives, targeting vulnerable groups such as young people, and those facing social exclusion. The ESF+ also plays a critical role in implementing the European Pillar of Social Rights, fostering a more inclusive and competitive labour market, while addressing challenges arising from the COVID-19 pandemic and the green and digital transitions.

In this study, we employ the spatial dynamic computable general equilibrium model RHOMOLO to assess the interim macroeconomic impact of the ESF+ funds for the whole of the programming period 2021-2027 and up to 13 years later, until 2040 (see also European Commission, 2025). We use the available financial allocations as of the end of 2024. The impact is measured at the Member State level and for the whole of the EU, illustrating an improvement in Member States' GDP by about 0.175% and employment prospects, while finding that cross-country disparities notably decline.

Interim impact assessments can complement in improving the ongoing implementation and adjustment of policies such as the ESF+ by estimating potential impacts, refining policy implementation, enhancing policy responsiveness and assessing its progress. This helps identify potential areas of success and shortcomings, allowing for timely adjustments and evidence-based decision-making. Our interim analysis shows country-by-country results that shed light on labour market adjustments and the extent of disparities in response to planned policy investments to improve labour market participation and labour productivity, thus enabling a more effective allocation of funds.

Extant studies that provide policy impact evaluations of the ESF have been performed using either an econometric or by employing macroeconomic modelling approaches. Following the former, Canova and Pappa (2025) employ an econometric approach, specifically instrumental variable Bayesian local projections, to uncover significant medium-term impacts in terms of regional multipliers with substantial heterogeneity across EU regions for the planned investments of the ESF+ 2021-2027. González-Alegre (2017) employs panel data techniques to track the efficiency of the ESF 1989-2010 in Spanish regions, finding that ESF transfers can lead to disincentives in local active labour market policies expenditure at the regional level. On the other hand, Biedka et al. (2022) using a spatial panel model with data from 2007 to 2015 reveal that funds augmenting human capital can have a substantial impact in Polish regions. Fusaro and Scandurra (2023) examine the period between 2007 and 2018 using instrumental variables, and estimate a positive employment impact of the ESF on population shares with lower-secondary and a negative on populations having upper-secondary education[1], and a uniformly positive effect on youth employment across all education levels.

Following a macroeconomic modelling approach, Sakkas (2018) employs the RHOMOLO model to estimate the interim impact of the 2014-2020 ESF allocations finding that the fund contributes positively to regional growth and employment. Casas et al. (2025) using a variant of the RHOMOLO model incorporating endogenous labour participation, conduct an ex-post evaluation of the 2014-2020 ESF, including the Youth Employment Initiative and REACT-EU funds to find substantial positive adjustments to regional GDP and employment as well as a long-term reduction in regional disparities and educational mismatches.

The remainder of the paper is structured as follows. Section 2 discusses the model employed for the interim impact assessment. Section 3 presents the data and the simulation strategy. Section 4 discusses the results. The last section concludes.

[1] These results are consistent also with the findings of the ex-post evaluation of the Fund for European Aid to the Most-Deprived 2014-2020 (De Quinto Notario, A., 2024).

2. The RHOMOLO model with endogenous labour participation

The RHOMOLO model is a dynamic spatial computable general equilibrium model based on the extant literature of CGE models. The model is calibrated with reference to a set of regional social accounting matrices for year 2017 and cover all NUTS2 regions of Europe accounting for input-output linkages, income distributions, and cross-country trade flows (García Rodríguez et al., 2025). For the purpose of this exercise, the model is aggregated to the Member State level, each comprising of ten economic sectors (using the NACE Rev.2 Industrial classification).

We assume that within countries, households consume a fixed proportion of their income, while firms maximise profits and produce value added according to a constant elasticity of substitution production function. The production function comprises allocations between the paid factors of production capital and labour, and an unpaid factor of production being public capital (Baxter and King, 1993). The value added produced is then blended with goods or services produced elsewhere, the mix of which is governed by a respective elasticity of substitution, in order to produce output of goods or services. Firms are able to export abroad their output of goods or services indexed as specific varieties belonging to a particular sector (and country) at a transport cost which is assumed to be of the iceberg type (Krugman, 1991)[2] and increases with distance. The matrix of transport costs across EU regions and Member States is obtained from Persyn et al. (2020 and 2023).

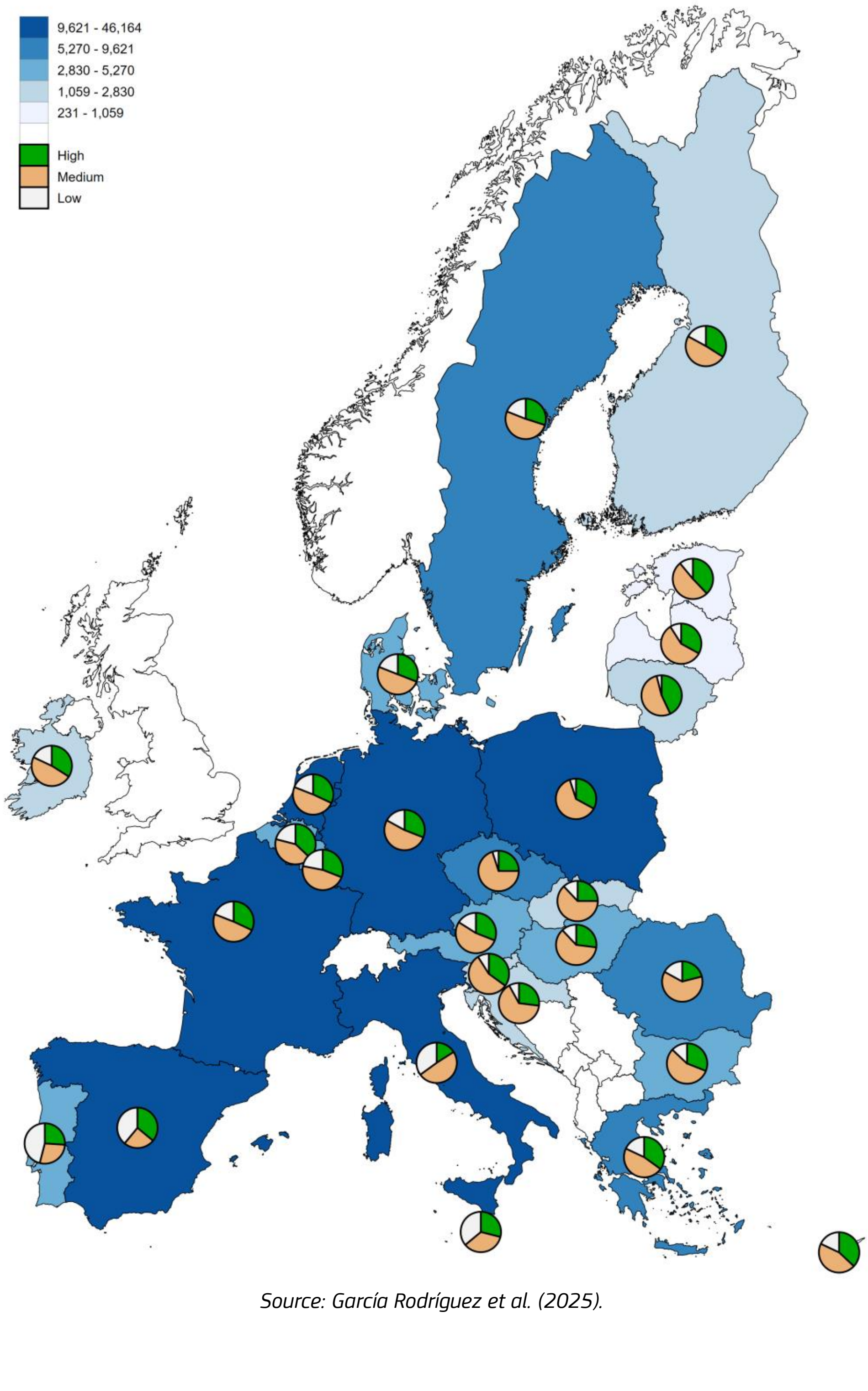
The model's labour force comprises three different categories of labour, indexed by each type's level of educational attainment as reported in the Labour Force Survey (LFS), and is based on the international standard classification of education (ISCED) for 2011. This allows us to construct the stock of labour per Member State and education level as per Figure 1. For each type of labour, wages are set by employing a wage curve as per Blanchflower and Oswald (1995), which enables that lower levels of unemployment increase workers' bargaining power, leading to an increase in real wages. In such a setting however, individual workers' decisions about their labour market status as well as their amount of supplied labour is ignored when considering reactions to economic or policy shocks, the ESF+ labour market policies being such an example. This rigidity is alleviated when incorporating into the model endogenous labour participation decisions of individuals, following Kleven and Kreiner (2006) and Boeters and Savard (2013). Hence prospective workers first decide the amount of labour hours they will supply in employment given they are employed. In the next step, they compare the utility of this outcome relative to the cost of participating (in other words, relative to receiving a benefit), and decide whether or not to enter the labour market.

The full mathematical description of the RHOMOLO model and its extension incorporating endogenous labour market participation decisions are presented in Lecca et al. (2018) and Christensen and Persyn (2022), respectively. The model is based on a reference year of calibration, or baseline, where all EU Member States are assumed to be in their steady state and is interpreted as the scenario in which no ESF+ funding is distributed to Member States (See also García Rodríguez et al., (2025) for the construction of the baseline). Policy shocks, represented as monetary injections that affect certain aspect(s) of a Member State's, or multiple Member States' economies, have the effect of changing the steady state, perturbing the endogenous variables of

[2] This implies that transport costs are converted in units of the transported good and expressed as the fraction of the goods shipped that is lost to transport.

the model, such as GDP, employment and levels of capital. The shock is transmitted across time through the policy itself, as it may span more than one periods according to an exogenous time profile of investments, or through a recursively dynamic process.

Figure 1. Labour force and education level distribution, thousands of individuals, RHOMOLO baseline



Source: García Rodríguez et al. (2025).

In this process, the per-period sequence of static equilibria are linked by the law of motion of the state variables, such as capital. This implies that economic agents are not forward-looking and their decisions are based solely on current and past information. The results are interpreted as deviations from the baseline, and where explicitly mentioned, the effect can be cumulative. The shocks are composed from a distribution of ESF+ funds supplied by the European Commission's Directorate-General for Employment, Social Affairs and Inclusion, which are discussed in the following section.

The RHOMOLO model is managed and developed by the Joint Research Centre (JRC). General equilibrium models like RHOMOLO rely on simplifying assumptions because of the inherent complexity of the phenomena they seek to reproduce. The model simulations analyse a policy in isolation in order to identify its effects, without taking into account its interaction with other initiatives, national or European, even if in the same field. There is also a lack of detailed information on the typology of operations supported and on the characteristics of the participants in the programmes. The modelling results should therefore be interpreted with the following limitations in mind: (i) modelling assumptions have been used to incorporate information on the amount of money spent in the programmes into the scenario simulations; (ii) different typologies of interventions have been grouped into broader categories and economic channels in order to keep the analysis tractable; (iii) the model uses programme-related data on expenditure and types of participants as inputs, but then has to rely on assumptions (based on the scientific literature and additional data) to translate these inputs into outcomes; (iv) the modelling results are affected by the variation in modelling parameters and assumptions used to calibrate the shocks, i.e. the way in which the interventions are introduced into the model.

The implication is that the results should only be interpreted as an interim analysis in the sense that the input data refer to the latest information on planned funds as per the policy programmes, not because the GDP, employment and other variables are those actually observed. Nevertheless, the work is informative for understanding the policy, in particular its transmission mechanisms, spatial distribution, macroeconomic cost-effectiveness and sustainability.

3. ESF+ policy interventions and simulation strategy

ESF+ funds are classified under three broad categories of intervention targeting employment, education and training, and social inclusion. Each category consists of a set of specific objectives (SO's) which describe the policy intervention. Each SO is then mapped to a RHOMOLO input shock based on the description, which affects the demand side of the model and the supply side of the model. Table 1 presents the input shock correspondence for each SO and the total amount of planned funds that are to be distributed amongst Member States. Figure 2 presents the distribution of funds across Member States and types of labour. As this is the maximum level of disaggregation of the data, the analysis is carried out at the Member State level, and results are reported accordingly.

The broad areas of intervention for employment and education and training constitute about 33% of the total allocated funds while social inclusion represent about 34% of the total. The employment broad area consists of policy interventions affecting exclusively labour supply, while education and training target exclusively the upgrading of labour productivity. Social Inclusion contains interventions targeting the improvement of labour supply mechanisms by about 63% and labour productivity by about 37%. For the purpose of this exercise, it is assumed that about 23.39% of the €132,558 million of the ESF+ funds are assigned to Low educated individuals, 48.69% to Medium and 27.92% to the High educated.[3] Funds are split across labour types depending on a simple population share criterion based on the LFS (see also Figure 1) as there is no further information about the allocation of funds across types of labour for each Member State.

Table 1. Correspondence between Specific Objectives (SOs) and RHOMOLO input shocks

Broad Area	RHOMOLO Model Shock	Specific Objective	Description of the intervention	Planned amount (€)	Demand-side effects	Supply-side effects
Employment	Labour Supply	ESO4.1	Access to employment and activation measures for all	29,517,254,656	Increase in government consumption	Increase in labour supply
	Labour Supply	ESO4.2	Modernising labour market institutions	1,675,424,640	Increase in government consumption	Increase in labour supply
	Labour Supply	ESO4.3	Gender balanced labour market participation	3,625,049,856	Increase in government consumption	Increase in labour supply

[3] These shares are obtained on the basis of historical data from the preceding programming period, taking shifts in thematic allocations since then into account.

	Labour Supply	ESO4.4	Adaptation of workers and enterprises to change	9,000,793,088	Increase in government consumption	Increase in labour supply
Education and Training	Labour Productivity	ESO4.5	Improving education and training systems	9,432,033,280	Increase in government consumption	Increase in labour productivity
	Labour Productivity	ESO4.6	Quality and inclusive education and training systems	20,994,058,240	Increase in government consumption	Increase in labour productivity
	Labour Productivity	ESO4.7	Lifelong learning and career transitions	12,839,029,760	Increase in government consumption	Increase in labour productivity
Social Inclusion	Labour Supply	ESO4.8	Active inclusion and employability	20,989,130,752	Increase in government consumption	Increase in labour supply
	Labour Supply	ESO4.9	Integration of third country nationals	1,171,164,032	Increase in government consumption	Increase in labour supply
	Labour Supply	ESO4.10	Intergration of marginalised communities such as Roma	943,029,312	Increase in government consumption	Increase in labour supply
	Labour Productivity	ESO4.11	Equal access to quality social and healthcare services	16,727,711,744	Increase in government consumption	Increase in labour productivity
	Labour Supply	ESO4.12	Social integration of people at risk	5,643,252,224	Increase in government consumption	Increase in labour supply
			Total	132,557,931,584		

Source: Directorate-General for Employment, Social Affairs and Inclusion. Note that the total planned funds of the ESF+ amount to €142,7 billion, but the funds planned for interventions for technical assistance and material support to the most deprived are excluded from this analysis.

Based on financial data as of December 2024, about half of the allocated funds are distributed to Germany (10% of the total), Spain (11.9%), Italy (20.1%) and Poland (10.9%), while about 31.8% targets Eastern European Member States and 43.8% is directed to southern European Member States (Cyprus, Greece, Italy, Malta, Portugal and Spain) as shown in Table 2. Concerning the broad areas of implementation, for the employment area nearly 50% of the allocated funds are distributed to Italy (24%), Spain (16%) and Poland (10%), while for education and training Germany (9.8%), Spain (8%), France (9%), Italy (18%), Poland (12%) and Portugal (9%), receive approximately 65% of the broad area funding. About 61% of the allocation of the social inclusion broad area funds are assigned to Germany (10%), Spain (12%), France (9%), Italy (19%), and Poland (11%) (see also Tables B.1., C.1., and D.1. of the Annex).

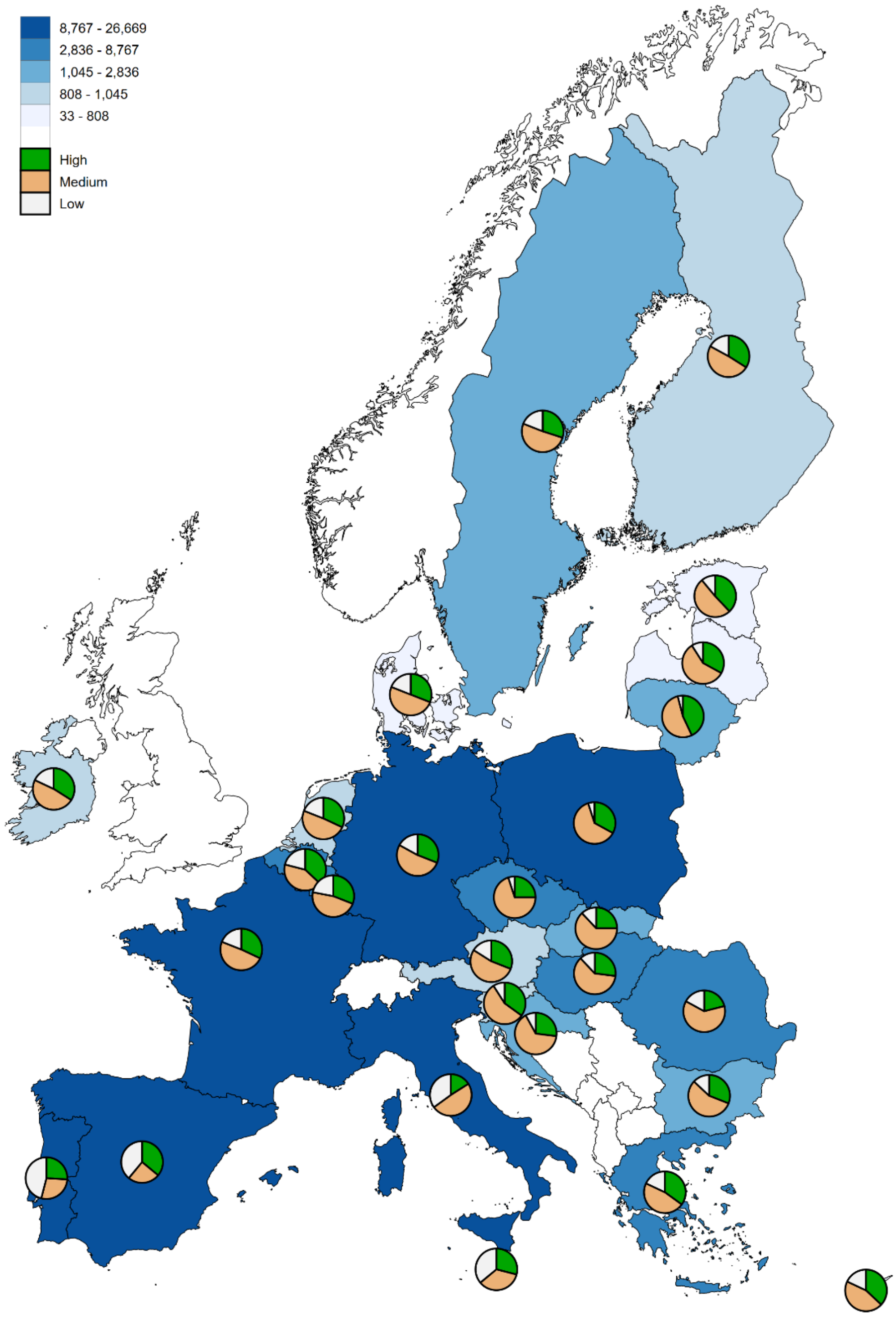
The funds, in addition to being distributed across levels of educational attainment and Member States, are allocated across time based on an assumed time profile which dictates the per period spending across the whole programming period (source: Directorate-General for Employment, Social Affairs and Inclusion).[4] Table 3 shows that approximately 88% of the funds are disbursed to Member States between 2025 and 2029, with about 47% paid out between 2025 and 2027 and 41% between 2028 and 2029. The disbursement of funds across time plays a significant role in determining the size of the deviations from the baseline. The earlier the total implementation the more significant the impact on key economic indicators such as GDP and employment, all else being equal, as firms and consumers have more per period optimising occurrences across the studied period.

The SO's defined in Table 1 allow the categorisation of funds into the model's exogenous input variables or shocks. SO's 4.1-4.4 and 4.8-4.10 and 4.12 are classified into a labour supply shock as they facilitate labour market participation, socio-economic integration and inclusion and modernisation of labour market institutions. The shock is assumed to increase the labour supply of Member States by investing in relevant training funded by the policy, on the basis of country-specific secondary education per cost per pupil (source: OECD (2024), "Education at a glance: Educational finance indicators"). The SO's are simultaneously assumed to increase government current expenditure for engaging in training and modernising labour market institutionscausing a temporary increase in demand for goods and services in a Member States' economy.

SO's 4.5-4.7 and 4.11 augment human capital and as a consequence are classified as a labour productivity shock. The interventions are implemented through additional years of schooling equivalents, again on the basis of a cost per person (based on country-specific tertiary education costs per student - source: OECD (2024), "Education at a glance: Indicators of education finance"). An additional year of schooling/training increases labour productivity according to a set of Mincer earnings function estimates (Card, 2001). The elasticity is based on country-specific estimates by Psacharaopoulos and Patrinos (2018a and 2018b). On the demand side, these expenditures are assumed to increase government current expenditure. The returns to labour supply and productivity are assumed to decay across time as workers gradually lose their learned skills as time elapses or are potentially exposed to new technologies. This decay rate is set to 5% per annum for both types of shocks, and as a consequence, the model in time would converge back to the baseline. As for the financing of the policy, each Member State pays a lump-sum contribution (non-distortionary, and reducing household disposable income as explained below) proportional to its GDP weight in the EU to mimic the contribution to the EU budget.

[4] For this time profile, financial absorption of ESF in the years 2014-2023 was used as a basis. It takes into account a slow start of the implementation and assumes that the implementation occurring (exceptionally) in 2023 is distributed into the last 4 years of the ESF+ implementation period.

Figure 2. ESF+ distribution of funds across Member States and types of labour (€ millions)



Source: Directorate-General for Employment, Social Affairs and Inclusion.

Table 2. Planned Allocation of ESF+ funds across Member States, by level of education

	Education Level				
Member State	**Low (€ 1000)**	**Medium (€ 1000)**	**High (€ 1000)**	**Total**	% of EU total
AT	149	501	295	945	0.71%
BE	598	1,178	1,060	2,836	2.14%
BG	348	1,582	886	2,816	2.12%

CY	56	144	119	319	0.24%
CZ	153	2,056	729	2,937	2.22%
DE	2,313	6,913	4,066	13,292	10.03%
DK	49	125	78	252	0.19%
EE	77	379	285	741	0.56%
EL	1,154	2,963	2,250	6,367	4.80%
ES	6,161	4,014	5,642	15,817	11.93%
FI	173	476	329	978	0.74%
FR	2,022	5,014	3,251	10,287	7.76%
HR	195	1,415	583	2,193	1.65%
HU	698	3,606	1,610	5,913	4.46%
IE	188	500	358	1,045	0.79%
IT	9,477	12,970	4,222	26,669	20.12%
LT	59	692	559	1,310	0.99%
LU	7	15	10	33	0.02%
LV	76	467	264	808	0.61%
MT	67	66	55	187	0.14%
NL	187	476	306	970	0.73%
PL	721	8,883	4,805	14,409	10.87%
PT	4,032	2,498	2,237	8,767	6.61%
RO	1,332	4,589	1,530	7,451	5.62%
SE	312	824	490	1,626	1.23%
SI	89	527	328	944	0.71%
SK	318	1,666	662	2,647	2.00%
Total	31,010	64,539	37,009	132,558	100.00%

Source: Directorate-General for Employment, Social Affairs and Inclusion and authors' calculations.

Table 3. Assumed time profile of ESF+ funds implementation, expressed as per annum proportions of total planned financing

Member State	2021	2022	2023	2024	2025	2026	2027	2028	2029
AT	0.00%	0.00%	0.00%	7.70%	18.20%	12.30%	25.50%	17.70%	18.60%
BE	0.00%	0.00%	1.50%	8.10%	15.10%	15.20%	21.10%	20.70%	18.20%
BG	0.00%	0.30%	4.70%	7.80%	15.20%	13.40%	18.20%	23.50%	17.00%
CY	0.00%	0.00%	4.10%	1.00%	25.00%	4.00%	1.20%	36.00%	28.60%
CZ	0.00%	0.80%	4.00%	7.90%	12.90%	14.20%	18.10%	21.90%	20.20%
DE	0.00%	2.80%	8.10%	11.20%	15.00%	15.20%	18.70%	15.00%	14.00%
DK	0.00%	0.60%	3.40%	6.20%	12.20%	11.70%	21.00%	23.30%	21.60%
EE	0.00%	0.60%	5.40%	9.20%	15.10%	17.10%	17.80%	14.50%	20.30%
EL	0.00%	3.10%	8.90%	8.30%	10.00%	12.90%	21.90%	17.40%	17.50%
ES	0.00%	0.00%	0.40%	7.40%	17.20%	14.40%	17.30%	22.50%	20.70%
FI	0.00%	1.40%	7.50%	11.70%	14.30%	14.60%	17.90%	16.70%	15.90%

FR	0.00%	0.10%	8.10%	8.00%	20.80%	10.10%	20.20%	16.20%	16.60%
HR	0.00%	0.00%	0.70%	2.80%	12.70%	13.10%	24.00%	28.00%	18.70%
HU	0.00%	0.00%	1.20%	8.80%	19.20%	16.30%	21.30%	20.70%	12.60%
IE	0.00%	0.00%	0.00%	0.00%	43.30%	20.60%	7.80%	29.20%	0.00%
IT	0.00%	0.30%	3.20%	7.20%	14.70%	13.80%	20.80%	23.10%	17.00%
LT	0.00%	0.30%	9.10%	7.10%	9.50%	14.40%	26.70%	19.20%	13.70%
LU	0.00%	0.00%	2.00%	6.40%	7.30%	5.60%	5.20%	67.30%	6.00%
LV	0.00%	0.00%	6.60%	8.80%	13.00%	14.90%	18.20%	18.60%	20.00%
MT	0.00%	0.00%	0.80%	2.60%	7.40%	4.80%	6.60%	32.70%	45.20%
NL	0.00%	0.00%	9.30%	20.30%	9.40%	9.60%	24.50%	14.40%	12.50%
PL	0.00%	0.40%	4.70%	8.40%	13.20%	16.70%	18.90%	21.20%	16.50%
PT	0.00%	6.90%	8.50%	10.60%	12.80%	11.70%	13.10%	16.60%	19.90%
RO	0.00%	0.00%	0.10%	0.50%	16.10%	16.10%	26.50%	20.10%	20.70%
SE	0.20%	1.00%	6.20%	11.20%	16.30%	14.40%	13.20%	16.50%	21.10%
SI	0.00%	0.00%	5.20%	-3.00%	26.90%	10.90%	16.10%	19.40%	24.40%
SK	0.00%	0.00%	4.40%	6.80%	10.10%	12.80%	26.00%	23.80%	15.90%

Source: Directorate-General for Employment, Social Affairs and Inclusion.

Given this setting, an investment that increases labour productivity or labour supply has the following effects. In each year of the policy implementation according to Table 3, each educational group within a Member State has to choose to enter the labour market or not. If they do so, they need to define the amount of labour hours they will supply. The condition for entering the labour market depends on the relationship between the wage they will receive and the amount of benefits each group receives when unemployed, modelled as transfers from the government. The improvement of human capital and the entry of additional people into the labour force, causes firms to re-optimise their production inputs between capital and labour. Simultaneously, they also re-optimise the allocation of labour inputs across Low, Medium or High-educated workers. A higher labour productivity increases wages, which further increase consumption. Conversely, more workers entering the labour force may cause the opposite effect by reducing wages, while lowering unemployment rates. On the demand side increased government current expenditure leads to higher levels of consumption and pushes prices upwards. Lastly, the policy is assumed to be financed through household taxation, which leads to a reduction in disposable household income. The repayment of the policy is assumed to be proportional to each Member State's contribution to the EU budget, leading to an allocation of net beneficiary Member States, which receive more funds than they are required to pay back, and net contributor Member States, which contribute more to the financing of the policy relative to what they receive.

4. Results

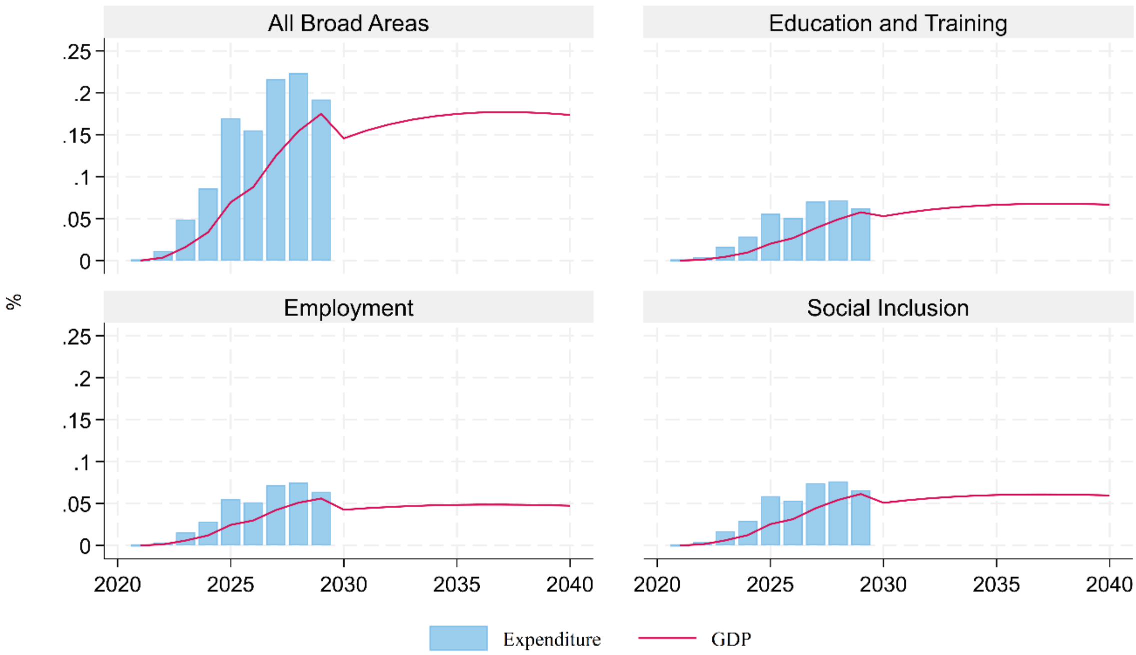
4.1. The ESF+ impact on GDP

The results of this section pertain to a simulation lasting 20 periods and cover the programming period 2021-2027 and up to 2040 incorporating all broad areas of Table 1 and each broad area separately.

Figure 3 shows the total estimated GDP impact measured at the EU level, overlaid on the cost of the policy for each broad area and in total. The GDP impact is measured in percentage deviations from the baseline EU GDP while the total funds of the policy are expressed as a percentage of EU GDP. Table 4 summarises these results for two years, in 2029 and 2040. In 2029, the last year of the enactment of the policy based on the time profile of Table 3, we observe the highest estimated GDP impact standing at +0.175% (+€21,028 million) compared to the baseline, or in other words, compared to a scenario without the policy. 10 years after the start of the policy in 2030, a lower percentage change is observed in the model due to the termination of the demand component of the policy, leading to a lower year-on-year level of consumption, but still 0.15% higher than baseline GDP. From 2031 onwards, the GDP impact is estimated to gradually increase as the now trained individuals, after undertaking training during the policy implementation, enter the labour force and/or employment. In 2040, the model predicts that the impact remains positive standing at +0.173% (+€20,879 million) compared to the absence of the policy. The implementation yields a long-term return to investment, also referred to as the cumulative GDP multiplier, of 2.28 euros for every euro invested.[5] Each broad area contributes to the overall through its expected impact on GDP depending on the type of the intervention, labour productivity or labour supply, and the allocated amounts.

[5] The multiplier is defined as the cumulated GDP change relative to the baseline in a particular year divided by the cumulative amount of the policy expended in the same year. For example in Table 4, the GDP multiplier is the quotient of the cumulative Impact (€ millions) by the cumulative policy expenditure (€ millions) columns, yielding the euro return for every euro of policy expended.

Figure 3. EU-wide GDP Impact (% deviations from baseline) and size of interventions (% of GDP)



Source: RHOMOLO simulations.

The employment broad area is expected to contribute about 27% to the overall GDP deviations from the baseline, or about 0.047% in 2040 according also to Figure 3. This translates to a long run multiplier of about 2 (see Table 4). The policy which only targets labour supply, generates positive and significant GDP returns, however these are slightly lower compared to the returns of the other two broad areas which contain also labour productivity interventions. The education and training broad area is expected to yield relatively higher long run GDP impacts across Member States compared to the employment broad area of intervention. In 2040 according to the model it yields approximately 39% of the overall GDP impact of the three Broad Areas, compared to 27% of the employment broad area while both interventions are approximately of equal size. This difference in GDP returns is due to higher levels of labour productivity in the production of output, leading to higher wages and levels of consumption. The effect is persistent across time as shown in Figure 3. The social inclusion broad area contains both a mix of policies and is slightly larger than the other two broad areas. The GDP impact is estimated to be slightly higher than the employment Broad Area and given the relatively low weight of the labour productivity interventions (37%), the GDP response is anticipated to be slightly lower than the education and training broad area. Figure 3 and Table 4 confirm this as the GDP impact at the end of the policy implementation in 2029 and in the long run, in 2040, stands at +0.06% higher compared to baseline GDP.

Table 4. EU-wide GDP impact, policy investments and cumulated GDP multipliers in 2029 and 2040

Broad Area	Year	GDP			Cumulative Policy expenditure (€ millions)	GDP Multiplier
		% Deviations from baseline	Annual impact € (Million)	Cumulative Impact (€ millions)		
All	**2029**	0.175%	21,028	79,904	132,558	0.60
	2040	0.173%	20,879	303,157	132,558	2.28
Employment	**2029**	0.056%	6,720	26,687	43,819	0.61
	2040	0.047%	5,688	88,784	43,819	2.03
Education and Training	**2029**	0.058%	6,924	24,927	43,265	0.58
	2040	0.067%	8,023	109,291	43,265	2.53
Social Inclusion	**2029**	0.061%	7,369	28,356	45,474	0.62
	2040	0.060%	7,167	105,257	45,474	2.31

Source: RHOMOLO simulations.

The impact of the policy is heterogeneous across Member States both in terms of GDP impact and in terms of multipliers. Table 5 shows that the long-run GDP multipliers across Member States tend to be large and positive, the size of which depends on the amount of funds received by the policy, the type of policies contained in each broad area and the amount of GDP generated by each Member State. Among other things, the size of the multiplier depends on whether a Member State is a net contributor or net beneficiary of the policy. For instance, a country receiving a small amount of funding but experiencing relatively large GDP gains can be characterised by large multipliers. Large GDP gains when policy expenditure is low in a Member State can be generated through spillovers (that is, economic effects generated by investments elsewhere). Spillovers are related to trade. Member States that engage more in exports to satisfy short-term increase in demand via increased government expenditures elsewhere, may experience large gains, as is the case with the long-run multipliers of Luxembourg and Ireland (receiving 0.02% and 0.79% of the total funding of the ESF+ respectively).

On the other hand, a high-income country that receives relatively little funding (as compared to their GDP) but contributes significantly to the financing of the policy may experience a negative GDP impact (and therefore a negative multiplier), such as Denmark in the short term (in 2029). This depends very much on the amount of funds each Member State receives, the GDP impact the funds generate exclusively from the application of the policy, and the amount of imports the receive to accommodate the short term increase in government consumption expenditures (the demand side of the shock). Small interventions generally have a small GDP response that comes exclusively from the policy. The short run horizon also does not allow firms to adjust all factors of production, only labour, affecting value added production. Higher imports coming from abroad, given the size of the policy funding can offset any small gains in GDP from the policy, all other factors constant, in the short run. In the long-run once all factors of production can be adjusted, and the demand side of the shock is extinguished, the structural part of the shock which increases supply, wages and consumption leads to positive GDP multipliers (this is what happens with Denmark, in fact).[6]

Focusing on the employment broad area, all Member States are expected to experience long-run returns on average of about 2 euro for every euro of the policy invested, slightly lower than the average multiplier observed from the combined three broad areas of intervention. This is due to the lack of labour productivity interventions in this broad area. The education and training broad area, which comprises labour productivity augmenting interventions yields relatively higher GDP impacts causing higher multipliers. On average they are about 2.5 euro for every euro invested (See Tables 4 and 5). Lastly, the social inclusion broad area yielding a multiplier of about 2.31 euro for every euro invested, is slightly higher than the employment broad area but lower than the education and training broad area, owing to the policy mix of both labour supply and labour productivity.

Table 5. Country-wide cumulated GDP multipliers in 2030 and 2040, by Broad Area and overall

	2029				**2040**			
	Broad Area				**Broad Area**			
Member State	**Employment**	**Education and Training**	**Social Inclusion**	**Total**	**Employment**	**Education and Training**	**Social Inclusion**	**Total**
AT	-0.84	0.38	0.23	0.20	1.93	3.02	2.54	2.75
BE	0.44	0.35	0.53	0.44	2.31	1.96	2.47	2.22
BG	0.41	0.34	0.37	0.37	1.03	0.89	0.96	0.96
CY	0.53	0.56	0.59	0.56	1.46	2.47	2.38	2.10
CZ	0.50	0.42	0.50	0.47	2.03	2.10	2.07	2.07
DE	0.74	0.74	0.76	0.74	3.19	4.21	3.30	3.55
DK		-0.18	-1.05	-0.97		2.04	2.32	1.54
EE	0.62	0.39	0.58	0.49	2.24	1.26	2.01	1.68
EL	0.78	1.11	1.00	0.93	1.51	3.76	3.01	2.54
ES	0.67	0.69	0.67	0.68	1.89	2.83	2.01	2.14
FI	0.59	0.40	0.59	0.53	2.74	2.56	2.89	2.72
FR	0.42	0.57	0.60	0.54	2.37	3.33	2.66	2.83
HR	0.54	0.52	0.53	0.53	1.68	2.08	1.97	1.91
HU	0.59	0.47	0.57	0.52	2.00	1.79	1.98	1.89
IE	-0.01	0.01	0.66	0.35	2.42	2.41	3.60	3.03
IT	0.68	0.64	0.66	0.66	2.07	2.47	2.32	2.27
LT	0.67	0.73	0.73	0.72	2.11	3.09	2.94	2.77

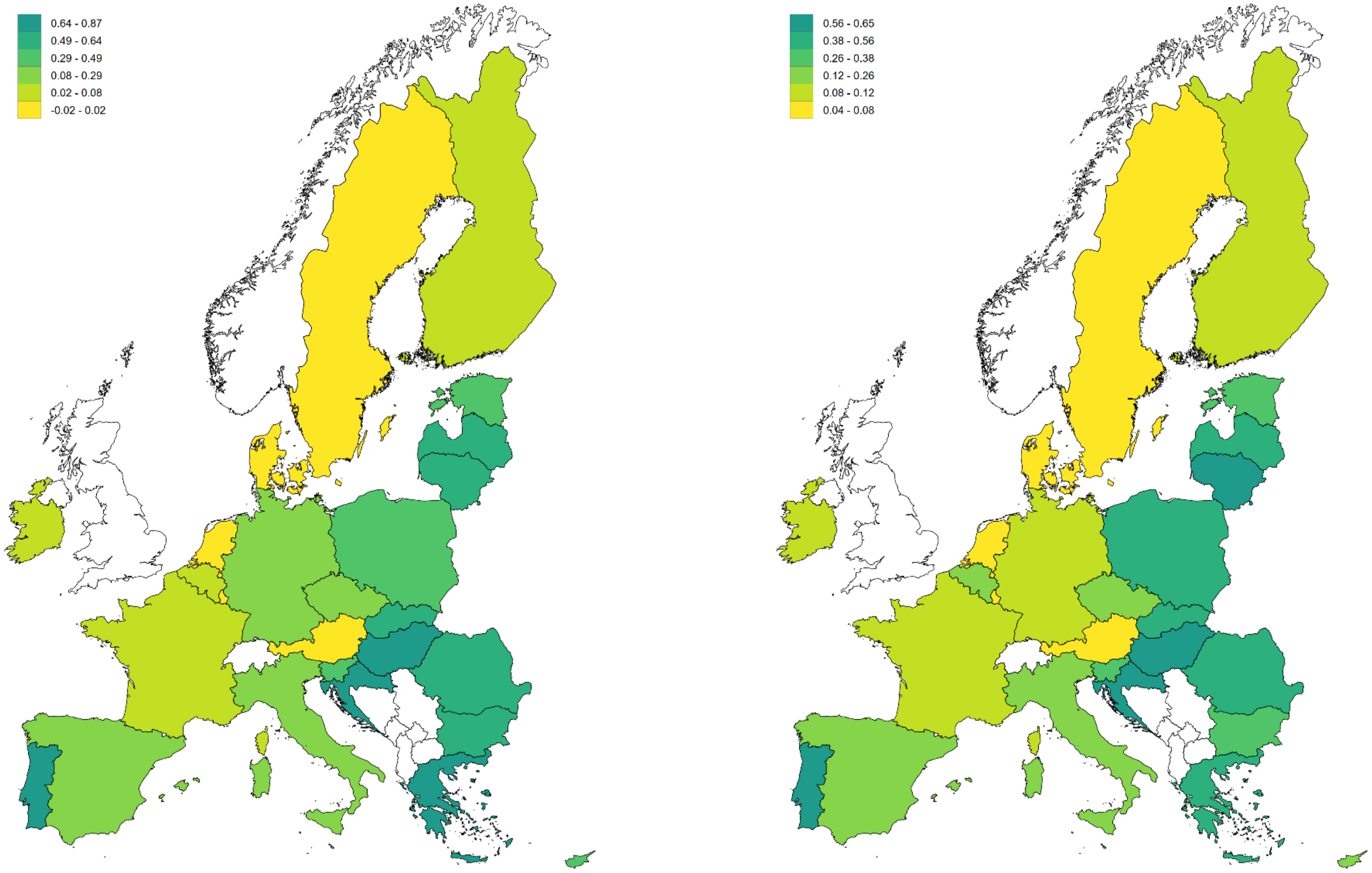
[6] For a thorough discussion on GDP multipliers and spillovers please refer to Barbero et al (2024).

LU	0.83	0.50	0.96	0.75	6.53	7.94	9.01	7.73
LV	0.59	0.70	0.67	0.67	1.83	2.91	2.40	2.48
MT	0.45	0.32	0.38	0.37	2.00	2.03	2.03	2.02
NL	0.37	0.01	0.40	0.28	2.78	2.96	3.12	2.95
PL	0.53	0.47	0.50	0.50	1.69	1.79	1.75	1.75
PT	0.67	0.74	0.71	0.72	1.71	2.41	2.11	2.14
RO	0.45	0.41	0.41	0.42	1.40	1.47	1.45	1.44
SE	0.05	-0.26	0.59	0.23	2.34	0.89	3.07	2.29
SI	0.60	0.53	0.57	0.57	2.18	2.43	2.32	2.31
SK	0.65	0.53	0.62	0.61	2.15	2.01	2.11	2.10

Source: RHOMOLO simulations.

Figure 4 shows the heterogeneous GDP impact in 2029 and 2040 confirming that both in the short- and in the long-run the GDP impact is higher in Eastern and Southern European Member States compared to the rest of Europe and range between +0.38% to +0.65% relative to the absence of the ESF+ policy.

Figure 4. GDP Impact (% deviations from baseline)



(a) 2029 *(b) 2040*

Source: RHOMOLO simulations.

4.2. The ESF+ impact on the labour force and employment

The ESF+ impact on the labour force is estimated to be positive and large. Table 6 and Figures 5 and 7 show that on aggregate, in 2029 about 688 thousand people are expected to enter into the EU labour force compared to the absence of the policy. In the long term this impact is found to be persistent as in 2040 the policy leads to +411 thousand individuals entering the labour force in that year.[7] The impact is positive for all Member States with the largest adjustments occurring in Southern European Member States, Poland and Germany. Proportional to the distribution of the labour force across labour types, nearly half of the individuals entering the labour force are Medium educated (49%), while the remainder are Low (22%) and High educated (29%). We observe little variation in the distribution of individuals across education levels as there is no further information about how the funds are distributed across labour types within Member States The change relative to the baseline across labour types is about equal to the mean change of 0.185%. The labour force adjustments by Member State and labour type are presented in Table A.1. of Annex A.

Changes in the labour force and in employment are for the most part a result of policies improving labour supply (SO's 4.1-4.4 and 4.8-4.10 and 4.12). In this case, Table 7 and Figure 6 reveal that about 500 thousand people are estimated to enter the labour market during the last year of the policy implementation, and about +300 thousand in 2040.[8] The distribution of new jobs is slightly skewed toward Medium and High educated as the long run change affects the Medium (+0.15%, +154 thousand) and High educated (0.15%, +93 thousand) more than the Low educated (+0.14%, +56 thousand). These results depend on the amount of allocated funds to each education group as well as the distribution of the population across labour types for each Member State. For example, as per Figure 1 Lithuania and Poland have a relatively low percentage share in Low educated populations. The employment impact of the policy reveals that the reaction of the Low educated in Poland and Lithuania based on Figure 8 is therefore less. In 2040, about 4% (1,500 new workers) of the total change in the number of new employees in Poland amounting to 40,900 persons are Low educated, while 60% (24,650 workers) are Medium educated and 36% (14,750 workers). In Lithuania 5% (210) of all new workers (+4,260) are Low educated, while 56% (2,400) are Medium educated and 39% (+1,650) are High educated. A detailed breakdown of changes in employment are presented in Table A.2. of Annex A.

On the contrary, Member States with a higher relative stock of Low educated such as Portugal show a higher proportion of the impact affecting the Low and Medium educated compared to the High educated, reinforced by proportionate distributions of policy funding. In Spain in 2040 for example, 30% (10,200) of all new workers (33,600) are Low educated, while 25% (8,400) are Medium and 44% (15,000) are High educated. This is quite different in comparison to Portugal, where 48% (7,800 workers) of all new workers (16,200) are Low educated, 28% (4,600) are Medium educated and 24% (3,800) are High educated.

The expected impact of the employment broad area, consisting exclusively of labour supply interventions, is particularly effective, as it contributes 61% of the total long run change in the labour force observed in Table 6. About 254 thousand of the total 411 thousand individuals

[7] These changes are annual relative to the baseline and not cumulative.

[8] These changes are annual relative to the baseline and not cumulative.

entering the labour force are due to this broad area of interventions. At the end of the policy programming period, when the highest impact is observed, this adjustment measures about 430 thousand individuals entering employment. The impact is proportional to the distribution of the labour force across education types, with most of the impact affecting Medium educated individuals. In terms of employment, the impact is proportional to the amount of the interventions allocated to each education group and the distribution of the population across education groups within each Member State. At the end of the policy implementation period, the intervention is expected to lead to about +282 thousand individuals entering employment and about +148 thousand in the 2040. These figures correspond to about 56% of the total change in employment from the three broad areas combined in 2029 and 49% of the total adjustment in 2040.

The education and training broad area does not target labour market institutions and has a negligible impact on the entry into the labour force. Table 6 shows that the short and long run changes amount to about 2 thousand individuals moving in or out of a labour force of 222 million. The policy also has a small and positive change in employment, as it does not necessarily lead to new jobs being created, but mainly making existing workers more productive. As such in the long run there are approximately 61 thousand jobs created in 2029, compared to the 282 thousand jobs created under the "Employment" Broad Area of intervention. In 2040 these figures amount to 61 thousand and 148 thousand jobs respectively.

Lastly, the social inclusion broad area is estimated to yield sizeable adjustments in labour force entry and in employment entry. About 155 thousand individuals enter the labour force, approximately 37% of the total adjustment of the three broad areas. Since the funding is proportional to the population, the Medium educated are the ones that benefit the most from the policy, followed by the high educated and low educated. In terms of employment, the policy contributes about 37% to the total employment impact of the three Broad Areas combined, again with the biggest adjustment affecting the Medium educated. Such changes in the labour force and employment come exclusively from the labour supply interventions, since the labour productivity shock as observed in the "Education and Training" Broad Area had negligible effects on labour market entry or employment, affecting mostly GDP instead. We conclude that the SOs promoting labour market integration and human capital upgrade show positive results across all types of labour and Member States, yet the intensity of the change depends on the distribution of education levels in the population and the allocation of funds across education groups. The responses of both the labour force and employment to the policy injection are steeper during the first 9 years of implementation compared to the following 11 years as in the former period both the supply and demand effects of the shocks are active. During the latter period, only the structural component pertaining to the supply side remains, resulting in positive responses but their magnitude declines across time due to depreciation of labour force entry incentives and workers becoming less productive across time by using skills that they learn in the present day.

Table 6. Change in the labour force relative to the baseline (number of persons)

Education Level		Low	Medium	High	Total
Year	Baseline / Broad Area	**47,775,394**	**109,446,318**	**65,188,710**	**222,410,416**
2025	**Employment**	+31,261 (0.07%)	+68,286 (0.06%)	+40,631 (0.06%)	+140,178 (0.06%)

	Education and Training	-635 (-0.00%)	-1,162 (-0.00%)	-1,909 (-0.00%)	-3,706 (-0.00%)
	Social Inclusion	+19,300 (0.04%)	+44,438 (0.04%)	+27,109 (0.04%)	+90,847 (0.04%)
	Total	+49,985 (0.10%)	+111,620 (0.10%)	+66,049 (0.10%)	+227,653 (0.10%)
2029	**Employment**	+95,624 (0.20%)	+211,016 (0.19%)	+123,248 (0.19%)	+429,888 (0.19%)
	Education and Training	+155 (0.00%)	-417 (-0.00%)	-1,394 (-0.00%)	-1,656 (-0.00%)
	Social Inclusion	+55,384 (0.12%)	126,182 (0.12%)	+78,536 (0.12%)	+260,102 (0.12%)
	Total	+151,179 (0.32%)	+336,781 (0.31%)	+200,532 (0.31%)	+688,492 (0.31%)
2035	**Employment**	+71,958 (0.15%)	+158,964 (0.15%)	+93,283 (0.14%)	+324,204 (0.15%)
	Education and Training	+308 (0.00%)	+1,025 (0.00%)	+459 (0.00%)	+1,792 (0.00%)
	Social Inclusion	+41,485 (0.09%)	+95,381 (0.09%)	+60,131 (0.09%)	+196,997 (0.09%)
	Total	+113,780 (0.24%)	+255,419 (0.23%)	+153,909 (0.24%)	+523,109 (0.24%)
2040	**Employment**	+56,423 (0.12%)	+124,789 (0.11%)	+73,087 (0.11%)	+254,298 (0.11%)
	Education and Training	+350 (0.00%)	+1,193 (0.00%)	+368 (0.00%)	+1,911 (0.00%)
	Social Inclusion	+32,637 (0.07%)	+75,164 (0.07%)	+47,109 (0.07%)	+154,910 (0.07%)
	Total	+89,418 (0.19%)	+201,158 (0.18%)	+120,574 (0.18%)	+411,150 (0.18%)

Source: García Rodríguez et al. (2025 – Baseline labour force), RHOMOLO simulations.

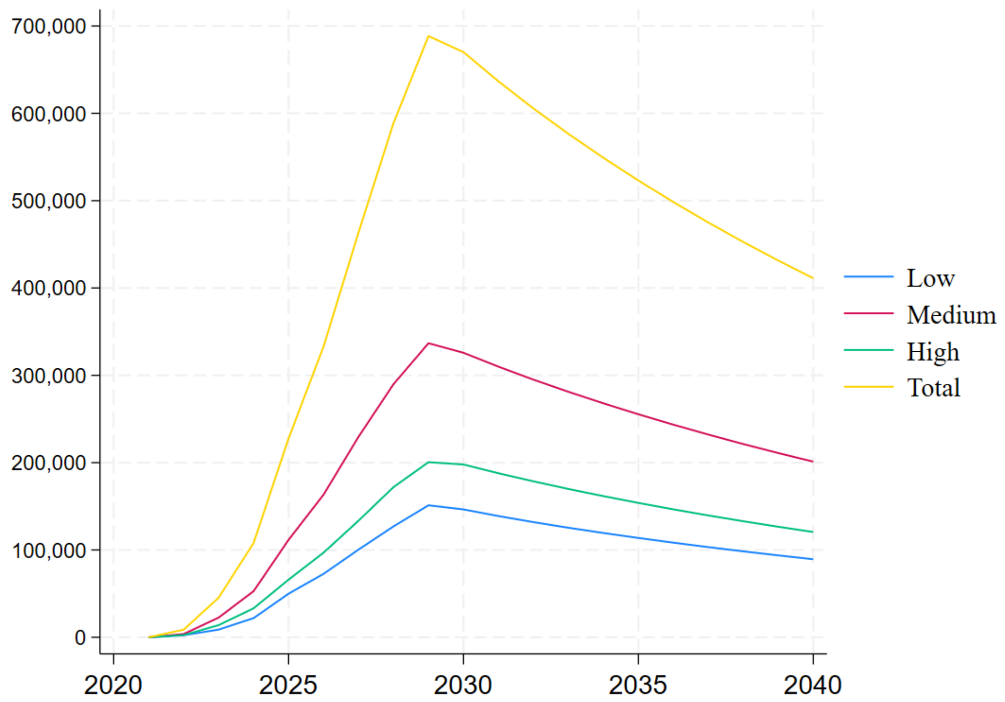
Table 7. Change in employment relative to the baseline (number of persons)

		Low	**Medium**	**High**	**Total**
Year	**Baseline** / **Broad Area**	**40,255,262**	**101,296,913**	**62,021,132**	**203,573,312**
2025	**Employment**	+30,052 (0.075%)	+70,466 (0.070%)	+49,178 (0.079%)	+149,696 (0.074%)
	Education and Training	+20,493 (0.051%)	+38,349 (0.038%)	+21,660 (0.035%)	+80,502 (0.040%)
	Social Inclusion	+26,588 (0.066%)	+62,882 (0.062%)	+41,620 (0.067%)	+131,091 (0.064%)
	Total	+77,386 (0.192%)	+171,924 (0.170%)	+111,950 (0.181%)	+361,261 (0.177%)
2029	**Employment**	+44,495 (0.111%)	+136,529 (0.135%)	+100,932 (0.163%)	+281,955 (0.139%)

	Education and Training	+15,245 (0.038%)	+27,234 (0.027%)	+18,757 (0.030%)	+61,236 (0.030%)
	Social Inclusion	+31,931 (0.079%)	+97,882 (0.097%)	+73,294 (0.118%)	+203,107 (0.100%)
	Total	+91,803 (0.228%)	+261,684 (0.258%)	+192,623 (0.311%)	+546,110 (0.268%)
2035	**Employment**	+22,945 (0.057%)	+83,794 (0.083%)	+58,677 (0.095%)	+165,416 (0.081%)
	Education and Training	+6,929 (0.017%)	+9,383 (0.009%)	+2,090 (0.003%)	+18,402 (0.009%)
	Social Inclusion	+16,151 (0.040%)	+57,982 (0.057%)	+40,383 (0.065%)	+114,515 (0.056%)
	Total	+45,997 (0.114%)	+150,994 (0.149%)	+101,012 (0.163%)	+298,003 (0.146%)
2040	**Employment**	23,237 (0.058%)	+74,867 (0.074%)	+49,453 (0.080%)	+147,557 (0.072%)
	Education and Training	+13,520 (0.034%)	+21,543 (0.021%)	+7,178 (0.012%)	+42,242 (0.021%)
	Social Inclusion	+19,453 (0.048%)	+57,993 (0.057%)	+36,394 (0.059%)	+113,840 (0.056%)
	Total	+56,209 (0.140%)	+154,402 (0.152%)	+93,009 (0.150%)	+303,620 (0.149%)

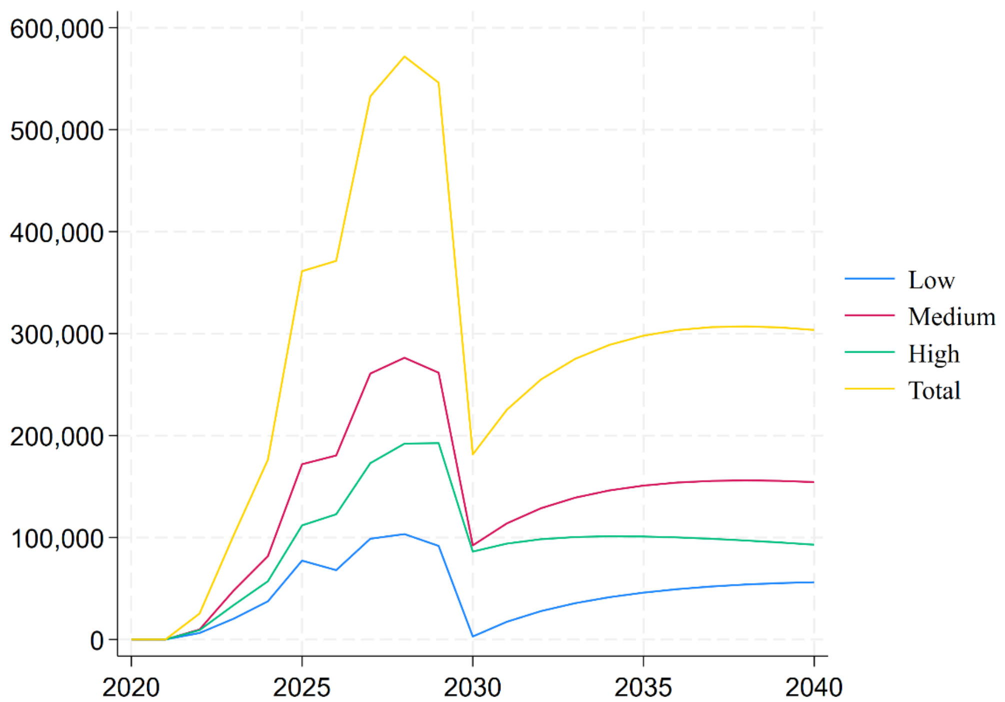
Source: García Rodríguez et al. (2025 – Baseline labour force), RHOMOLO simulations.

Figure 5. Annual labour force changes (number of persons)



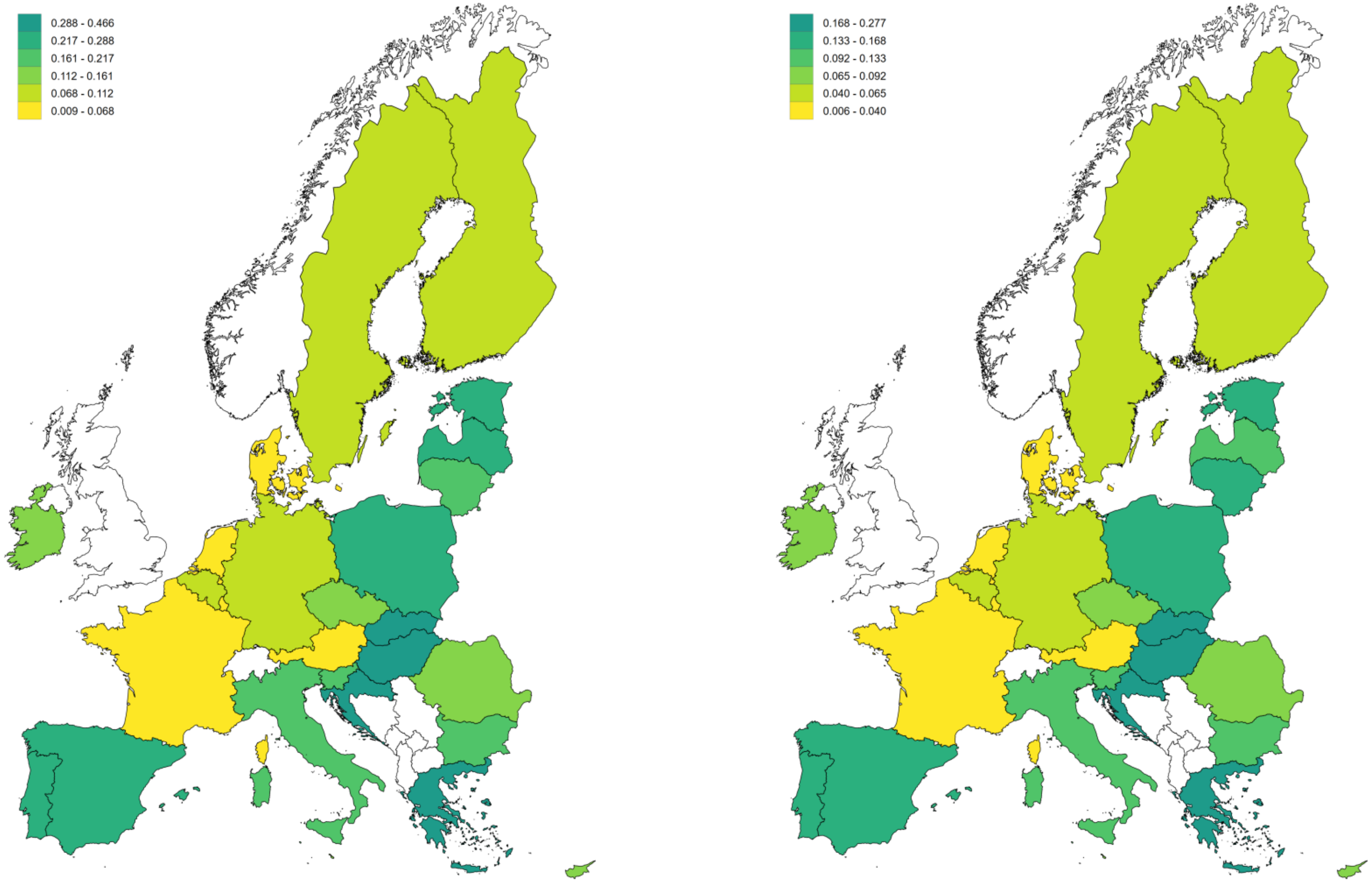
Source: RHOMOLO simulations.

Figure 6. *Annual employment changes (number of persons)*



Source: RHOMOLO simulations.

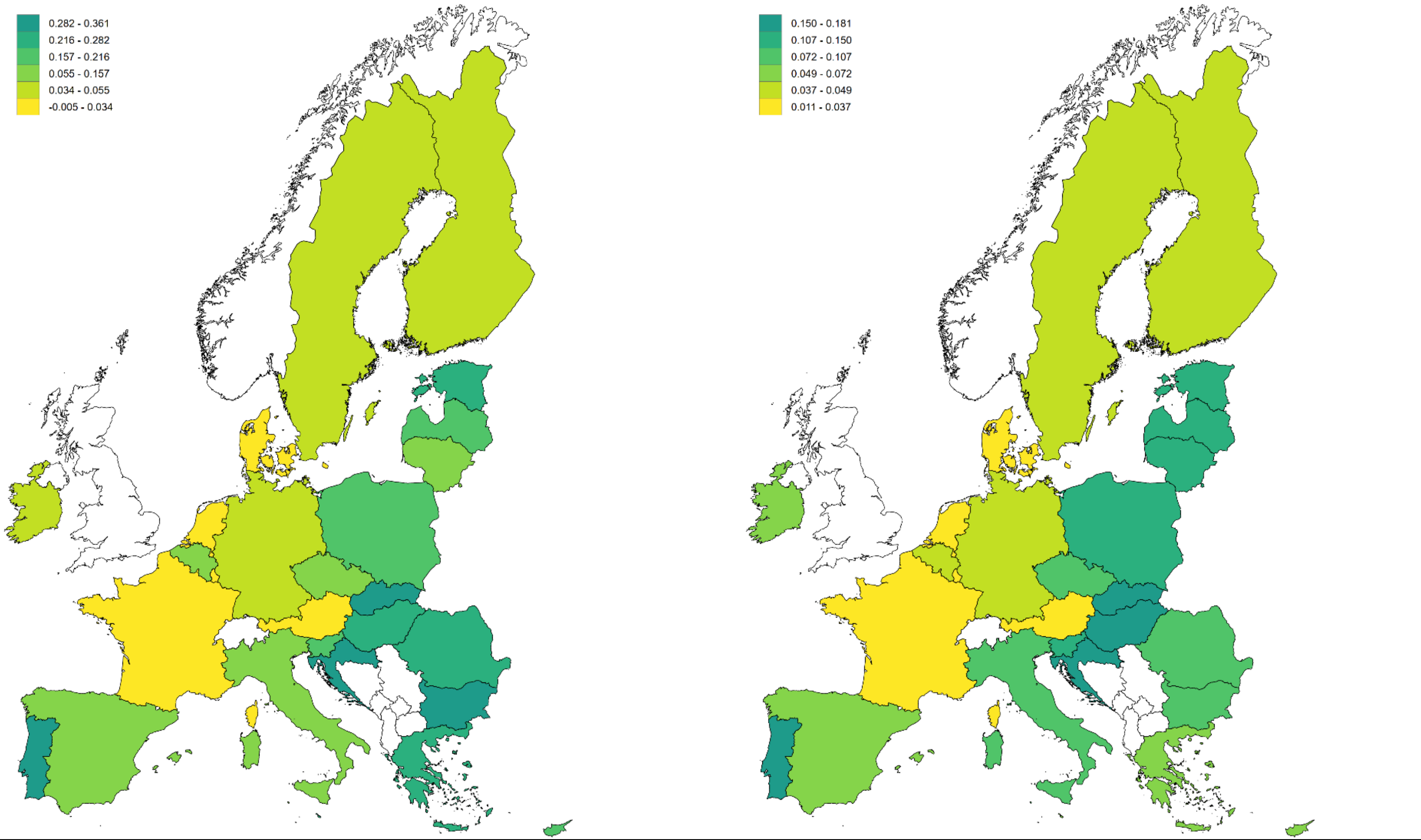
Figure 7. *Change in labour force expressed as % of the population*



(a) 2029 *(b) 2040*

Source: RHOMOLO simulations.

Figure 8. *Change in employment in 2040, expressed as % of the population*



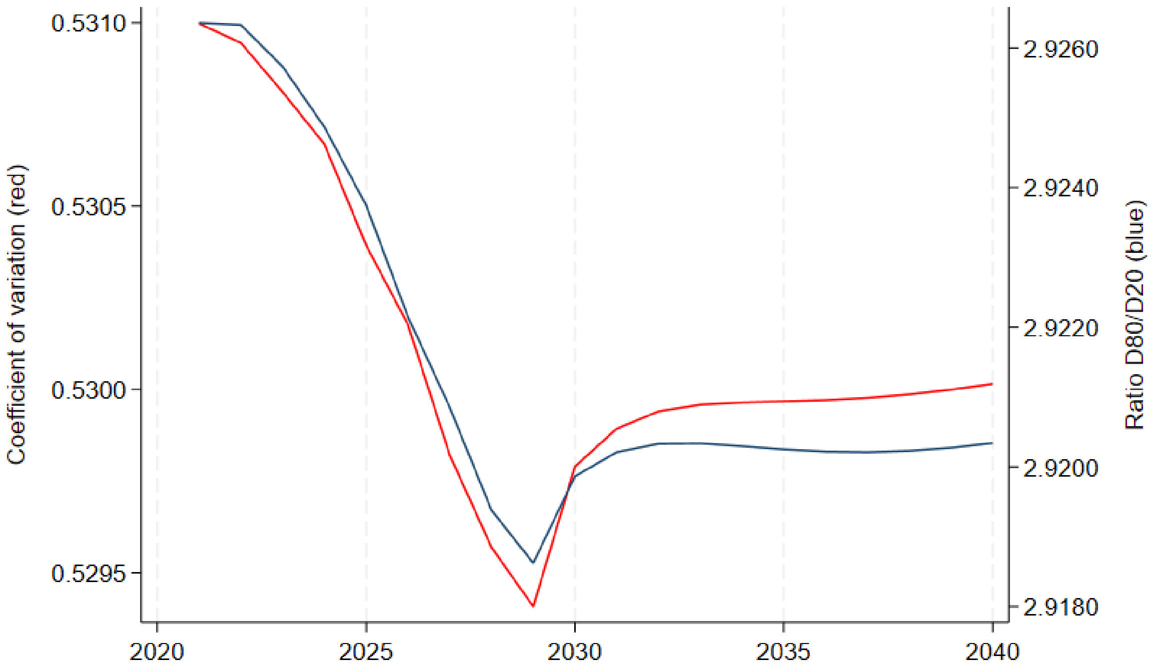
(a) 2029 *(b) 2040*

Source: RHOMOLO simulations.

4.3. The ESF+ impact on cross-country disparities

We employ a set of indicators to shed light on the role of ESF+ interventions in reducing cross-country disparities. Figure 9 shows the percentage change in the coefficient of variation on the left axis, and the percentage change in the interpercentile ratio of the 80th to the 20th percentile of GDP per capita of the EU.[9] These two measures show how the variation in incomes changes over time, with more variation leading to greater income disparities and vice versa. We observe a reduction in both measures of disparities as incomes gradually become more homogeneous, implying a reduction in the higher percentile incomes, an increase in the lower percentile incomes, and a smaller standard deviation. After year 2029, the decline is halted and remains relatively stable at about -0.23% compared to the baseline levels of the coefficient of variation and 80/20 ratio.

Figure 9. *Coefficient of Variation and ratio 80/20 of GDP per capita (% deviations from baseline)*



Source: RHOMOLO simulations.

Table 8 shows the percent change in per capita income for the bottom and top deciles and median income of Member States in 2040 relative to the baseline, that are the outcome of the policy implementation. The corresponding tables for each broad area are listed in Appendices B, C and D. On average, the policy is estimated to lead to an increase of lower incomes relative to the baseline,

[9] The coefficient of variation is defined as the ratio of the standard deviation of regional GDP relative to the mean regional GDP per capita. A declining ratio implies less variation and a more homogeneous level of GDP per capita that converges toward the mean and vice versa. The interpercentile ratio is the GDP per capita of a high percentile divided by a low percentile. We present the 80th/20th ratios.

a relatively stable median income and a decline of the top 10% of incomes. As discussed in the previous section, this adjustment comes from interventions that improve labour productivity. As firms optimise their labour input, education groups with higher relative productivity may be preferred compared to the high education group. The demand for more productive low or medium labour can lead to higher real wages compared to the high educated, especially in regions with higher shares of low and medium educated workers, leading to a decline in high decile incomes compared to an increase in low decile incomes. This reaction leads to a relatively more homogeneous income distribution compared to the baseline, confirming the decline in dispersion observed in Figure 9.

Table 8. Change in the income distribution in 2040 (% deviations from baseline)

Member State	Bottom 10%	Median (50%)	Top 10%
AT	0.02%	-0.01%	0.00%
BE	0.02%	0.00%	-0.02%
BG	0.00%	0.04%	-0.09%
CZ	0.02%	0.01%	-0.08%
DE	0.01%	0.00%	0.00%
DK	0.01%	0.00%	-0.01%
EL	0.07%	-0.11%	-0.14%
ES	0.03%	0.01%	-0.06%
FI	0.01%	0.01%	-0.02%
FR	0.01%	0.00%	0.00%
HR	0.02%	0.00%	-0.02%
HU	0.18%	0.13%	-0.54%
IE	0.00%	0.03%	-0.03%
IT	0.07%	-0.02%	-0.05%
LT	0.1%	0.00%	-0.1%
NL	0.01%	0.00%	-0.01%
PL	0.03%	-0.06%	0.02%
PT	0.05%	-0.09%	0.03%
RO	0.05%	0.03%	-0.16%
SE	0.02%	0.01%	-0.04%
SI	0.02%	0.00%	-0.02%
SK	0.12%	0.05%	-0.22%

Source: RHOMOLO simulations.

Table 9. Impact on the Theil Index (% deviations from baseline)

	Year	Baseline value	2029	2040
Employment	**Within**	0.014	-0.08%	-0.02%
	Between	0.052	-0.26%	-0.12%

	Overall	0.066	-0.22%	-0.10%
Education and Training	**Within**	0.014	-0.10%	-0.05%
	Between	0.052	-0.30%	-0.22%
	Overall	0.066	-0.26%	-0.18%
Social Inclusion	**Within**	0.014	-0.09%	-0.03%
	Between	0.052	-0.26%	-0.17%
	Overall	0.066	-0.22%	-0.14%
Total	**Within**	0.014	-0.27%	-0.11%
	Between	0.052	-0.82%	-0.51%
	Overall	0.066	-0.70%	-0.42%

Source: RHOMOLO simulations.

Another confirmation of less disparities in incomes per capita across the EU stemming from the interventions is the decline in the Theil index[10] of Table 8 at the end of the policy implementation in 2029 and 20 years after the start of the programme, in 2040. Both the between- and within-country components of the index decline, implying that disparities within Member States are reduced, as well as disparities across Member States.

The change in employment across education groups stemming from the funding leads to changes in the macroeconomic education mismatch (Kiss and Vandeplas, 2015, and Vandeplas and Thum-Thysen, 2019).[11] The overall impact shown in Figure 10 is the outcome of the labour supply interventions that affect employment as shown in Tables 7 and A.2 of the Annex (Figures B.4, C.4 and D.4 of the Annex show the change in the mismatch for each of the broad areas of intervention). Due to the agnostic distribution of funds across education groups and assuming an allocation of funds that is proportional to the populations of the education groups across Member States, we

[10] The index is calculated as: $Theil = \frac{1}{N}\sum_i^N S_j \, \frac{y_{ij}}{\bar{y}} \ln\left(\frac{y_i}{\bar{y}}\right) + \frac{1}{M}\sum_i^M S_j \, \ln\left(\frac{y_i}{\bar{y}}\right)$, where the first term of the formula represents the within part of the decomposition and is the weighted averages of the Theil index of each Member State. The second term is the between component of the Theil index and represents the component of disparities that depends on disparities across countries. S_j are weights and are computed as the ratio between the country average of income per capita, y, and its EU average.

[11] For each Member State, the indicator of macroeconomic educational mismatch (MEM) is defined as the relative dispersion of employment rates across three population groups with different educational attainment:

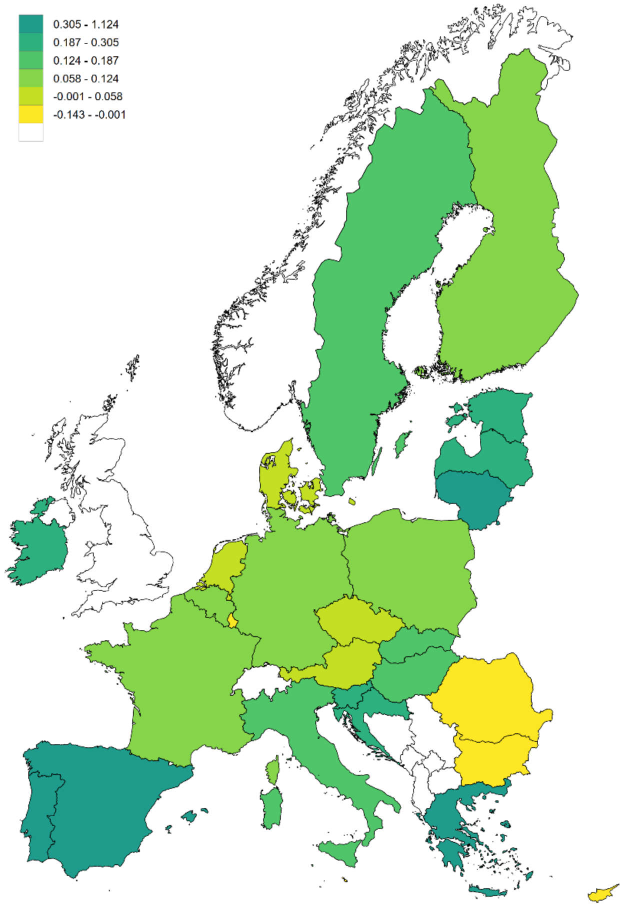
$$MEM = \sum_{i=L,M,H}\left|\frac{E_i}{E} - \frac{P_i}{P}\right| = \frac{1}{e}\sum_{i=L,M,H}\left|\frac{P_i}{P}(e_i - e)\right| \quad (1.1)$$

where E_i denotes employed persons with educational level i (either low, medium or high) of Member State; P_i is the respective working age population; and e_i is the respective employment rate. Variables with no index denote the corresponding aggregate country variables.

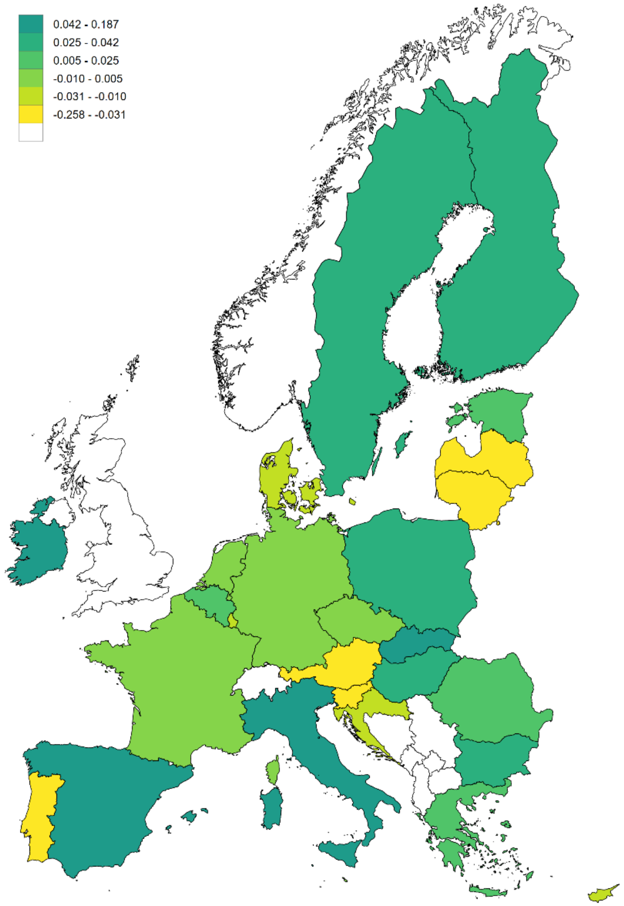
observe that the change in the mismatch is specific to a Member State's population distribution across education types.

For Member States with a large share of Low and/or Medium educated populations we observe a decline in the mismatch as there are relatively more Low or Medium educated workers entering employment and their group specific employment rate converges to the Member State's employment rate. In the long run in 2040, this can be observed with North European Member States such as Germany, the Netherlands and Denmark. Despite a high share of High educated to Low educated in Lithuania, Latvia and Poland, the interventions that mostly affect the Medium and Low educated, are estimated to lead to a decline in the education mismatch in 2040. The changes in the mismatch in South European Member States tend to be the opposite. Except for Portugal which consists of a high share of Low educated compared to High educated, leading to a decrease in the long-run educational mismatch, Member States such as Greece, Italy and Spain, experience an increase in the long-run educational mismatch.

Figure 10. *Change in the education mismatch (% deviations from baseline)*



(a) 2029



(b) 2040

Source: RHOMOLO simulations.

The three broad areas contribute to reducing disparities in different ways. The employment broad area is effective in reducing disparities as suggested by the reduction in the coefficient of variation and the interpercentile ratio of the 80th to the 20th percentile of GDP per capita in Figure B.3. Table 8 shows also a reduction in the Theil index of GDP per capita. However, the percentage decline is less than the overall decline of the three combined broad areas as the employment broad area interventions generate relatively lower per capita income compared the other two broad areas which also improve labour productivity and lead to higher levels of per capita income.

The employment broad area is also effective in making the income distribution in the long run relatively more uniform, as an improvement of the incomes of about 0.01% of individuals at the bottom 10% of the GDP per capita of the population in the baseline can be observed. For the populations receiving the median GDP per capita per Member State we see a negligible change on average across Member States in 2040 compared to the baseline income distribution, while for the highest decile of the income distribution we observe a reduction of about -0.02% in GDP per capita in 2040 compared to the baseline.

Lastly, the policy increases labour market mismatches throughout the EU. This happens because of the disproportional change in employment for the Medium and High educated populations which increases the dispersion of employment. Of the 148 thousand jobs created approximately, about 50% concern Medium educated and 33% High educated, and only 15% are Low educated jobs, which is roughly proportional to the distribution of funds across education groups within countries as shown in Table B.1. Different allocations of funding, more skewed toward improving labour market and employment access to the Low educated would lead to a reduction in education mismatches across time.

We conclude that the employment broad area yields a more than proportionate adjustment in labour force entry and in employment as approximately one third of the total policy amount generates more than 60% of the overall change in labour force entry and about +49% change in employment in the long run. The GDP returns are slightly lower compared to the other two broad areas, however they are sizeable, leading to an overall reduction in as incomes per capita tend to become more uniform across countries, as well as across education groups within countries.

The education and training broad area leads to a reduction of disparities due to the relatively higher amount of GDP per capita it generates. As shown in Table 8, the Theil Index reduces by about -0.04% in 2029 and -0.08% in 2040 compared to the employment broad area. The coefficient of variation and the interpercentile ratio imply even less variation in incomes compared to the employment broad area. The bottom 10% of the population in terms of GDP per capita sees their income increased by about +0.01% in 2040 compared to the baseline, while the top 10% experiences a decline in income of about 0.03%. As the policy causes smaller but positive changes in employment, the changes in the education mismatch relative to the baseline levels of education mismatch tend to be small (Figure C.4). Compared to the employment broad area interventions, the education and training broad area yields higher GDP returns from its policy implementation and leads to negligible changes in labour force entry and small adjustments in employment because of the lack of measures facilitating labour market entry. The higher incomes attributed to higher levels of labour productivity lead to relatively larger declines in disparities, with negligible effects on education mismatch.

The social inclusion broad area which combines both labour supply and labour productivity interventions improves disparities and contributes about 38% to the total effect of the three broad areas combined, as observed in the Theil index changes relative to the baseline and the coefficient of variation. Incomes adjust positively for the population groups with the lowest 10% per capita income relative to the baseline and negatively for the top 10% of income per capita. The policy leads also to declines in labour market mismatches in Romania, Bulgaria, Portugal and Greece, due to the relatively higher positive impact on low educated employment or the relatively lower impact on high educated employment.

5. Conclusions

We provide an interim macroeconomic impact assessment of three broad areas of the 2021-2027 ESF+ containing Specific Objectives designed to enhance human capital, social inclusion, and labour market initiatives, amounting to €132,557 million. We use the spatial dynamic general equilibrium model RHOMOLO, modified to include endogenous decisions of individuals about their labour market participation and intensity, to model the Specific Objectives as policy shocks affecting the labour supply and labour productivity across a 20-year period after the initiation of the policy.

We estimate that the policy has positive impacts on Member States' GDP, which is persistent across time, and amounting to about 0.17% at the end of the policy in 2029 and in 2040 on average. Across countries, the impact is heterogeneous and depends on the amount and type of interventions each country receives and implements as well as the distribution of the labour force across education types. The largest positive changes in GDP are documented in Eastern and Southern European Member States, which are also the main (net) beneficiaries of the ESF+ policy. The ESF+ interventions are estimated to generate positive returns to the policy investment, which at the EU level in 2040 stands at about 2.3 euro for every euro of the policy invested. The employment broad area consisting of specific objectives that improve labour market and employment entry contribute about 2.03 euros for every euro invested. The education and training equivalent value stands at 2.52 and it is higher as it consists exclusively of specific objectives that improve labour market productivity. The social inclusion broad area comprises a mix of labour productivity and labour supply specific objectives and yields a multiplier of 2.31. Since the shares of the policy interventions to the total amount are about the same, 33% of the total for the employment and education and training broad areas and 34% for social inclusion, we see that most of the effect on GDP change comes from the education and training broad area, whilst most of the labour market and employment adjustments stems for the other broad areas.

Interventions that aim to improve labour market access lead to a long run increase of the individuals entering the labour market of about 411 thousand, and policies that improve labour productivity for each type of labour, lead to a long run increase in employment of about 303 thousand new workers. The policy reactions mask considerable heterogeneity across countries and labour types. We observe that Eastern European Member States implement interventions that affect mostly the Medium and High educated because of the small proportion of Low educated workers. In Portugal and Italy the situation is opposite as most of the policy is enacted upon Low and Medium educated workers. Other Member States such as Greece or Spain have larger reactions of the High educated compared to the Low educated.

These responses lead to an unambiguous reduction in cross-country and within-country disparities as the dispersion of per capita GDP declines across time. The results shown herein hinge on the modelling assumptions and the planned versus actual allocation of funds across Member States. Therefore, the contents of this report should be considered as an approximation of the policy response by capturing the macroeconomic trend.

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Annexes

Annex A: Supplemental data

Table A.1. Change in the labour force in 2040, by education type and Member State (number of persons and share of total number of persons)

Member State	Low	Medium	High	Total
AT	129 (+0.11%)	613 (+0.54%)	393 (+0.35%)	1,135
BE	1,167 (+0.19%)	2509 (+0.42%)	2,325 (+0.39%)	6,001
BG	1,073 (+0.11%)	5315 (+0.57%)	2,969 (+0.32%)	9,357
CY	101 (+0.17%)	266 (+0.45%)	223 (+0.38%)	590
CZ	441 (+0.05%)	6471 (+0.72%)	2,127 (+0.24%)	9,038
DE	5,976 (+0.16%)	19426 (+0.53%)	11,403 (+0.31%)	36,805
DK	64 (+0.20%)	160 (+0.49%)	100 (+0.31%)	324
EE	214 (+0.10%)	1121 (+0.51%)	878 (+0.40%)	2,213
EL	4,703 (+0.17%)	12266 (+0.45%)	10,268 (+0.38%)	27,237
ES	22,833 (+0.37%)	15790 (+0.25%)	23,628 (+0.38%)	62,251
FI	557 (+0.17%)	1627 (+0.49%)	1,140 (+0.34%)	3,324
FR	4,973 (+0.18%)	13130 (+0.49%)	8,866 (+0.33%)	26,968
HR	606 (+0.08%)	4805 (+0.64%)	2,073 (+0.28%)	7,484
HU	2,520 (+0.12%)	12983 (+0.63%)	5,079 (+0.25%)	20,582
IE	664 (+0.16%)	1941 (+0.47%)	1,494 (+0.36%)	4,098
IT	26,386 (+0.35%)	36966 (+0.49%)	12,055 (+0.16%)	75,406
LT	178 (+0.04%)	2140 (+0.54%)	1,659 (+0.42%)	3,976
LU	12 (+0.23%)	22 (+0.42%)	18 (+0.34%)	52
LV	207 (+0.08%)	1445 (+0.57%)	868 (+0.34%)	2,519
MT	122 (+0.35%)	125 (+0.36%)	104 (+0.30%)	352
NL	514 (+0.19%)	1342 (+0.49%)	884 (+0.32%)	2,740
PL	2,352 (+0.05%)	31377 (+0.61%)	17,316 (+0.34%)	51,046
PT	7,815 (+0.46%)	4809 (+0.28%)	4,465 (+0.26%)	17,088
RO	2,959 (+0.16%)	11198 (+0.62%)	3,980 (+0.22%)	18,137
SE	809 (+0.17%)	2523 (+0.52%)	1,505 (+0.31%)	4,838
SI	235 (+0.09%)	1423 (+0.56%)	879 (+0.35%)	2,537
SK	1,810 (+0.12%)	9365 (+0.62%)	3,876 (+0.26%)	15,051
Total	89,418 (+0.22%)	201,158 (+0.49%)	120,574 (+0.29%)	411,150

Source: RHOMOLO simulations.

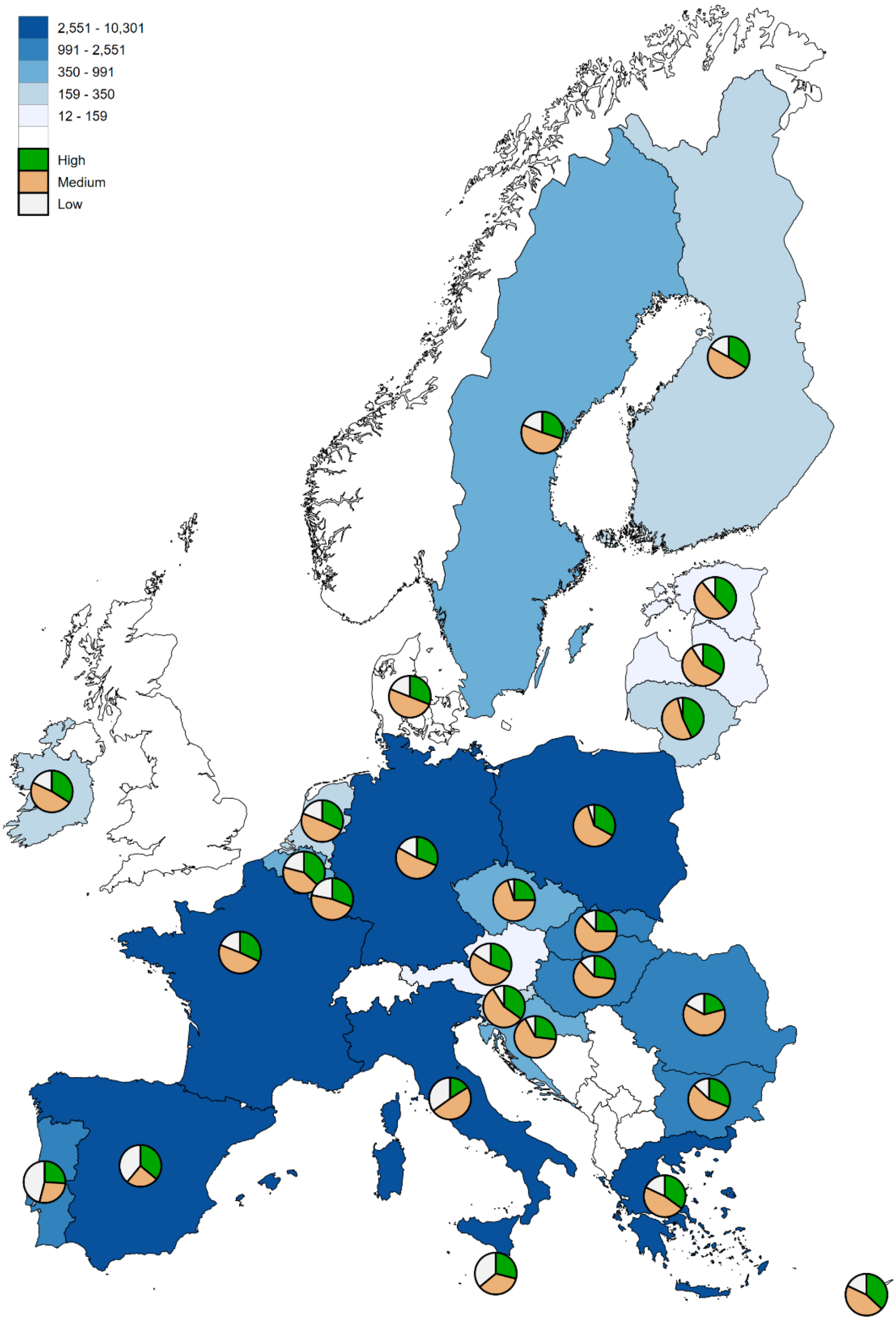
Table A.2. Change in employment in 2040, by education type (number of persons and share of total number of persons)

Member State	Low	Medium	High	Total
AT	396 (+0.19%)	1,101 (+0.53%)	576 (+0.28%)	2,073
BE	825 (+0.17%)	2,022 (+0.42%)	2,005 (+0.41%)	4,852
BG	601 (+0.08%)	4,359 (+0.58%)	2,521 (+0.34%)	7,482
CY	96 (+0.18%)	252 (+0.46%)	197 (+0.36%)	545
CZ	419 (+0.05%)	6,342 (+0.72%)	2,076 (+0.23%)	8,837
DE	6,489 (+0.17%)	20,224 (+0.53%)	11,707 (+0.30%)	38,420
DK	155 (+0.24%)	311 (+0.47%)	189 (+0.29%)	655
EE	175 (+0.09%)	955 (+0.50%)	763 (+0.40%)	1,894
EL	988 (+0.18%)	2,503 (+0.44%)	2,143 (+0.38%)	5,633
ES	10,214 (+0.30%)	8,406 (+0.25%)	15,018 (+0.45%)	33,637
FI	350 (+0.14%)	1,232 (+0.48%)	966 (+0.38%)	2,548
FR	4,436 (+0.18%)	12,050 (+0.48%)	8,407 (+0.34%)	24,893
HR	542 (+0.08%)	4,231 (+0.66%)	1,639 (+0.26%)	6,412
HU	1,672 (+0.09%)	11,454 (+0.65%)	4,594 (+0.26%)	17,719
IE	391 (+0.14%)	1,269 (+0.45%)	1,173 (+0.41%)	2,834
IT	13,749 (+0.31%)	22,224 (+0.50%)	8,268 (+0.19%)	44,241
LT	210 (+0.05%)	2,403 (+0.56%)	1,651 (+0.39%)	4,264
LU	42 (+0.29%)	66 (+0.45%)	39 (+0.27%)	147
LV	245 (+0.09%)	1,558 (+0.59%)	844 (+0.32%)	2,647
MT	111 (+0.35%)	112 (+0.35%)	94 (+0.30%)	317
NL	667 (+0.20%)	1,603 (+0.49%)	994 (+0.30%)	3,264
PL	1,522 (+0.04%)	24,655 (+0.60%)	14,744 (+0.36%)	40,921
PT	7,828 (+0.48%)	4,598 (+0.28%)	3,821 (+0.24%)	16,247
RO	2,825 (+0.17%)	10,340 (+0.62%)	3,410 (+0.21%)	16,575
SE	600 (+0.14%)	2,243 (+0.53%)	1,360 (+0.32%)	4,204
SI	244 (+0.10%)	1,463 (+0.57%)	839 (+0.33%)	2,546
SK	415 (+0.04%)	6,427 (+0.65%)	2,971 (+0.30%)	9,813
Total	56,209 (+0.19%)	154,402 (+0.51%)	93,009 (+0.31%)	303,620

Source: RHOMOLO simulations.

Annex B. Results for the employment broad area

Figure B.1. ESF+ distribution of funds across Member States and types of labour (€ millions)



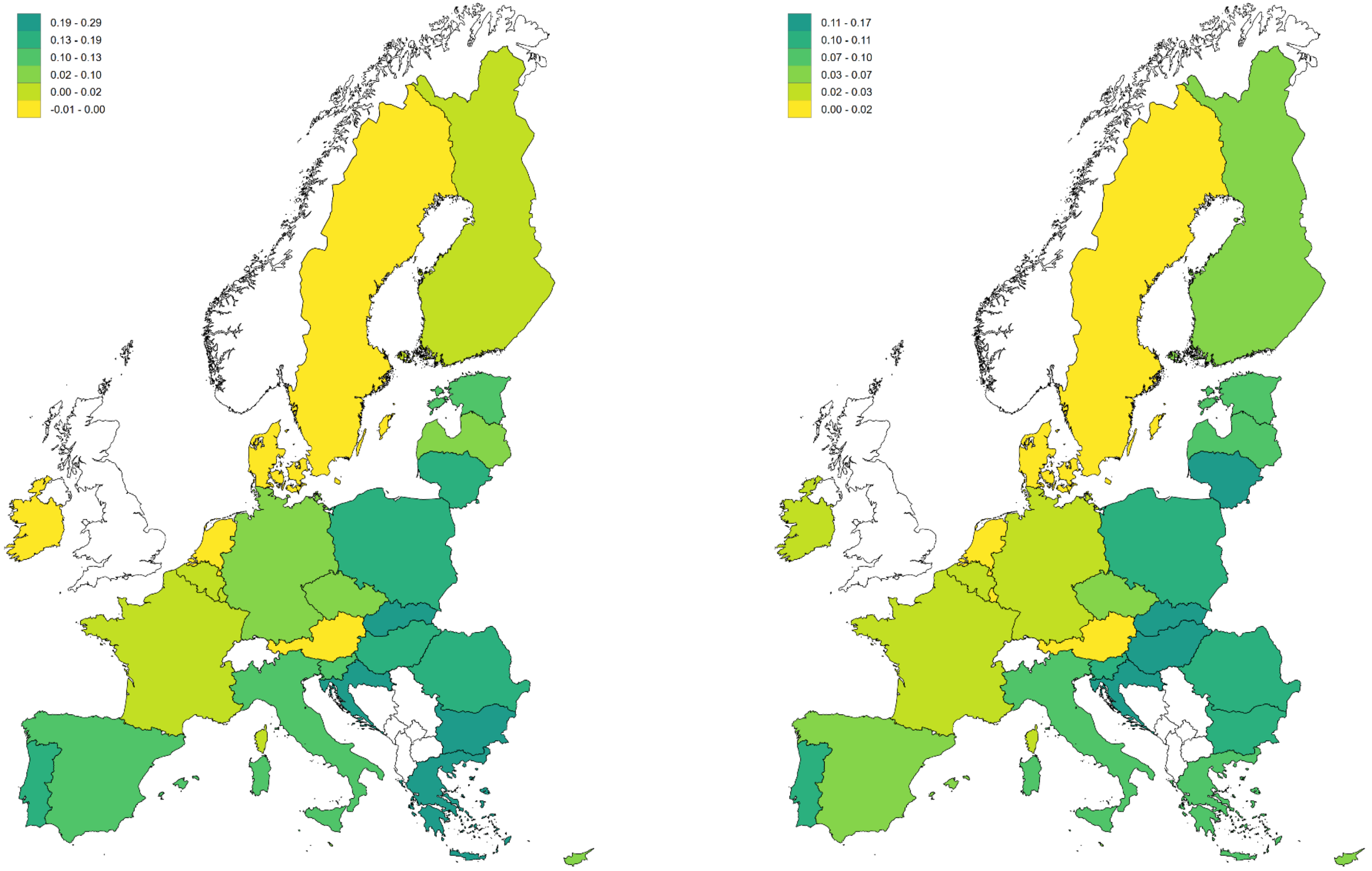
Source: Directorate-General for Employment, Social Affairs and Inclusion.

Table B.1. Planned Allocation of ESF+ funds across Member States, by level of education, thousands

	Education Level				
Member State	**Low (€ 1000)**	**Medium (€ 1000)**	**High (€ 1000)**	**Total**	% of EU total
AT	15	51	30	96	0.22%
BE	143	282	254	680	1.55%
BG	122	556	312	991	2.26%
CY	18	47	39	104	0.24%
CZ	44	589	209	841	1.92%
DE	757	2,262	1,331	4,350	9.93%
DK	-	-	-	-	0.00%
EE	14	69	52	135	0.31%
EL	504	1,295	984	2,783	6.35%
ES	2,690	1,752	2,463	6,905	15.76%
FI	59	162	112	332	0.76%
FR	502	1,243	806	2,551	5.82%
HR	66	481	198	746	1.70%
HU	150	773	345	1,268	2.89%
IE	34	92	66	192	0.44%
IT	3,661	5,009	1,631	10,301	23.51%
LT	15	176	142	332	0.76%
LU	3	6	4	12	0.03%
LV	15	92	52	159	0.36%
MT	14	14	12	41	0.09%
NL	68	172	110	350	0.80%
PL	228	2,812	1,521	4,562	10.41%
PT	997	618	553	2,168	4.95%
RO	399	1,373	458	2,230	5.09%
SE	69	184	109	363	0.83%
SI	27	160	100	288	0.66%
SK	125	655	260	1,041	2.38%
Total	10,739	20,927	12,152	43,819	100.00%

Source: Directorate-General for Employment, Social Affairs and Inclusion and authors' calculations.

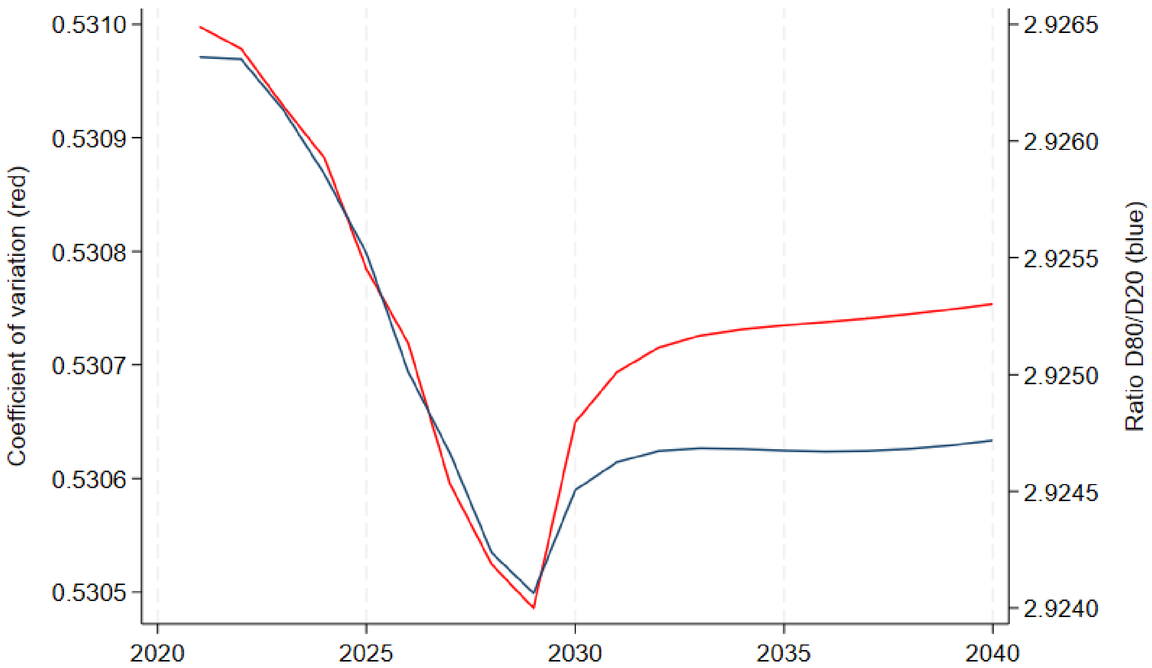
Figure B.2. GDP Impact (% deviations from baseline)



(a) 2029 *(a) 2040*

Source: RHOMOLO simulations.

Figure B.3. Coefficient of Variation and ratio 80/20 of GDP per capita (% deviations from baseline)



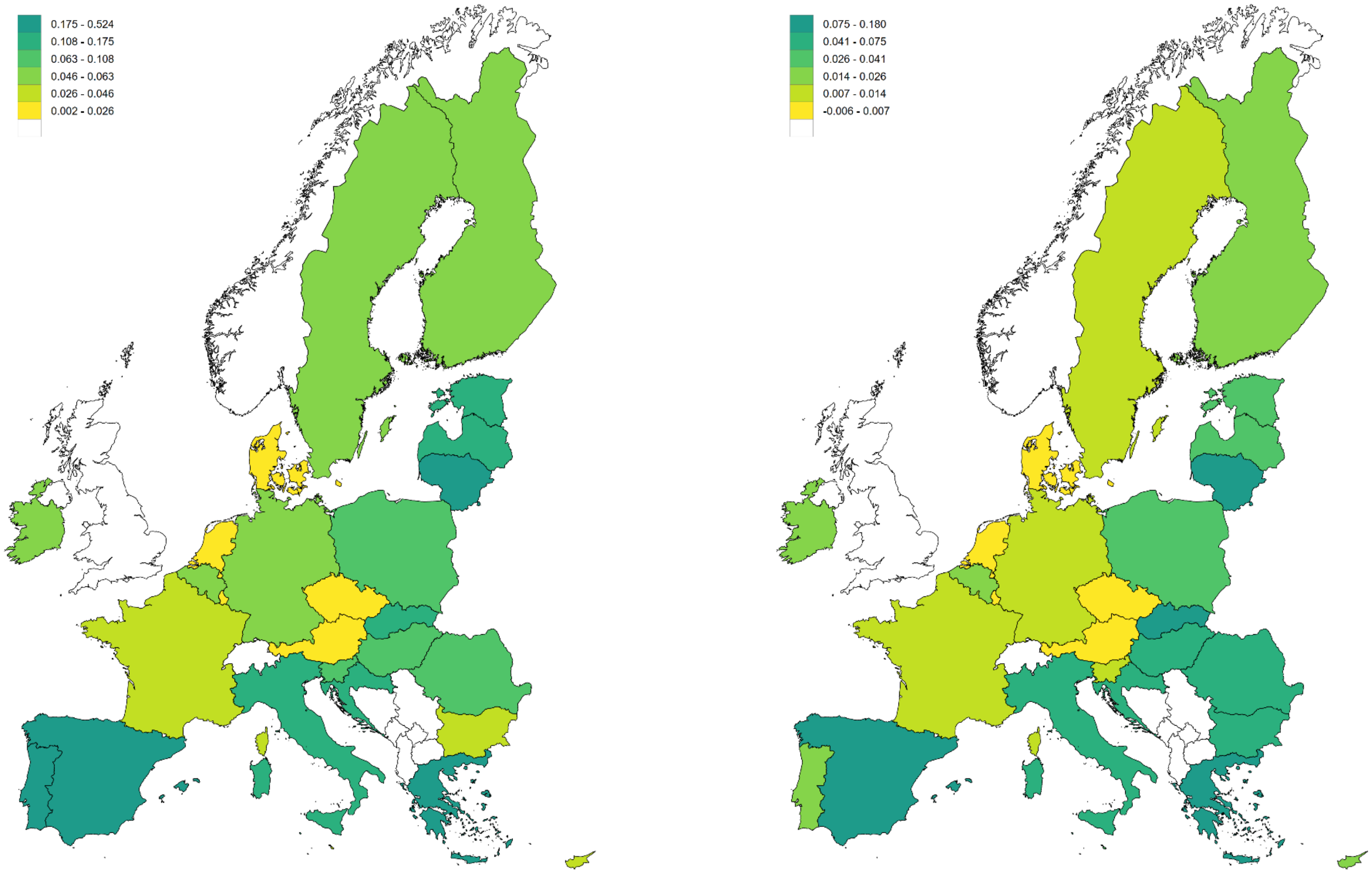
Source: RHOMOLO simulations.

Table B.2. Change in the income distribution in 2040 (% deviations from baseline)

Member State	Bottom 10%	Median (50%)	Top 10%
AT	0.004%	-0.002%	0.001%
BE	0.007%	-0.001%	-0.006%
BG	0.001%	0.014%	-0.03%
CZ	0.005%	0.003%	-0.024%
DE	0.005%	0.001%	0.00%
DK	0.002%	0.00%	-0.002%
EL	0.024%	-0.04%	-0.033%
ES	0.011%	0.004%	-0.038%
FI	0.003%	0.004%	-0.005%
FR	0.003%	0.001%	0.00%
HR	0.004%	0.00%	-0.004%
HU	0.04%	0.031%	-0.121%
IE	-0.001%	0.009%	-0.008%
IT	0.025%	-0.005%	-0.019%
LT	0.018%	0.00%	-0.018%
NL	0.002%	0.00%	-0.002%
PL	0.009%	-0.02%	0.007%
PT	0.009%	-0.023%	0.008%
RO	0.016%	0.007%	-0.047%
SE	0.006%	0.004%	-0.011%
SI	0.007%	0.00%	-0.007%
SK	0.048%	0.02%	-0.088%

Source: RHOMOLO simulations.

Figure B.4. Change in the education mismatch (% deviations from baseline)

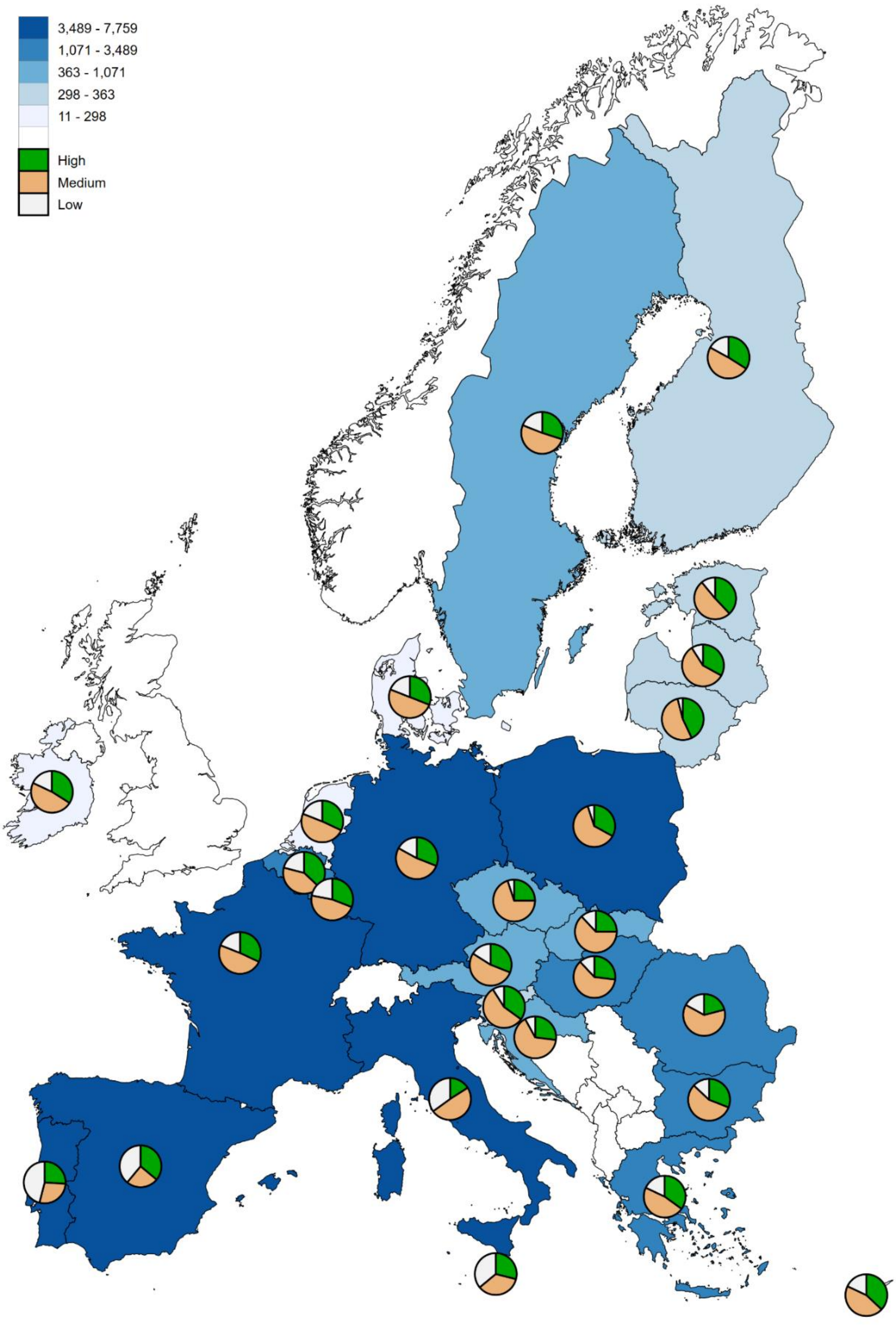


(a) 2029 *(a) 2040*

Source: RHOMOLO simulations.

Annex C. Results for the education and training broad area

Figure C.1. ESF+ distribution of funds across Member States and types of labour (€ millions)



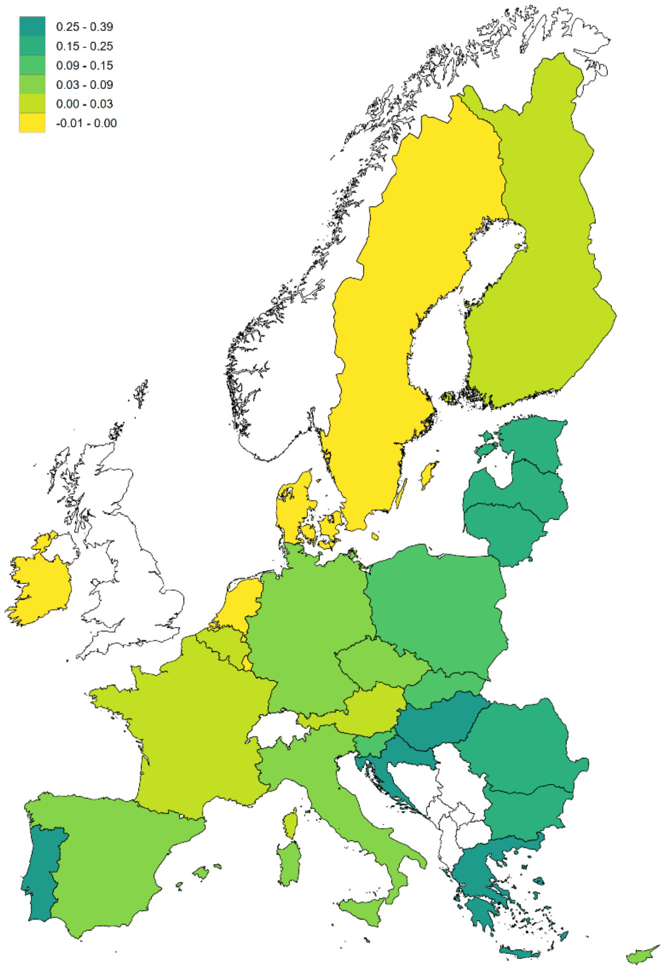
Source: Directorate-General for Employment, Social Affairs and Inclusion.

Table C.1. Planned Allocation of ESF+ funds across Member States, by level of education, thousands

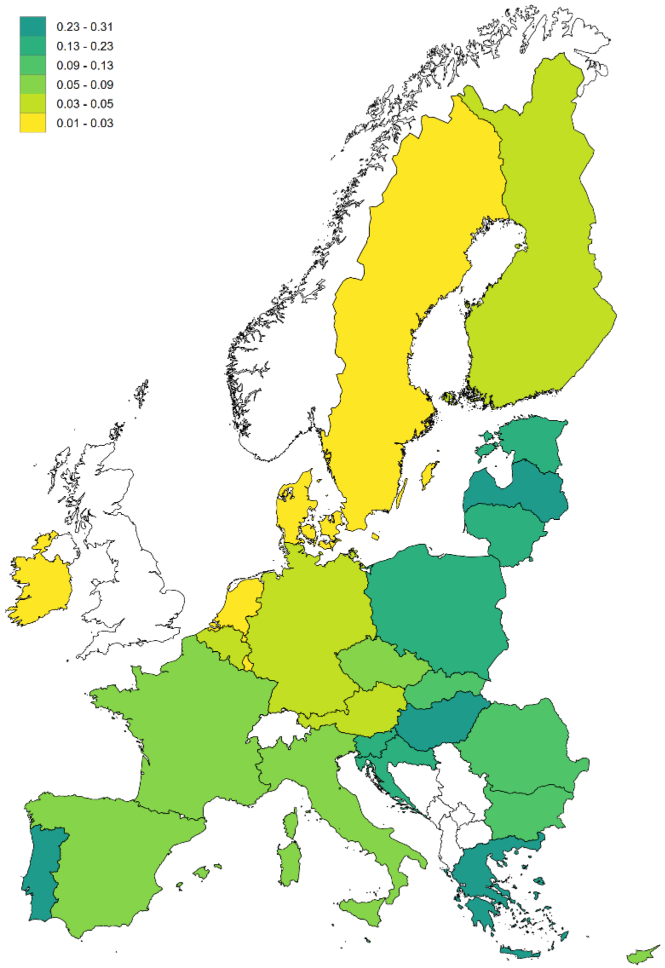
	Education Level				
Member State	**Low (€ 1000)**	**Medium (€ 1000)**	**High (€ 1000)**	**Total**	% of EU total
AT	86	290	171	547	1.26%
BE	246	485	437	1168	2.70%
BG	132	602	337	1071	2.47%
CY	13	34	28	76	0.17%
CZ	56	749	266	1070	2.47%
DE	743	2219	1305	4267	9.86%
DK	36	91	57	184	0.43%
EE	38	186	139	363	0.84%
EL	295	756	574	1625	3.76%
ES	1359	885	1244	3489	8.06%
FI	59	162	112	332	0.77%
FR	755	1872	1214	3841	8.88%
HR	64	468	193	725	1.68%
HU	367	1897	847	3110	7.19%
IE	54	143	102	298	0.69%
IT	2757	3773	1228	7759	17.93%
LT	16	186	150	352	0.81%
LU	2	5	3	11	0.03%
LV	29	176	100	305	0.70%
MT	32	31	26	89	0.21%
NL	52	132	85	269	0.62%
PL	247	3046	1648	4941	11.42%
PT	1720	1066	954	3740	8.64%
RO	407	1403	468	2277	5.26%
SE	87	231	138	457	1.06%
SI	32	186	116	334	0.77%
SK	68	356	142	565	1.31%
Total	9751	21431	12083	43265	100.00%

Source: Directorate-General for Employment, Social Affairs and Inclusion and authors' calculations.

Figure C.2. GDP Impact (% deviations from baseline)



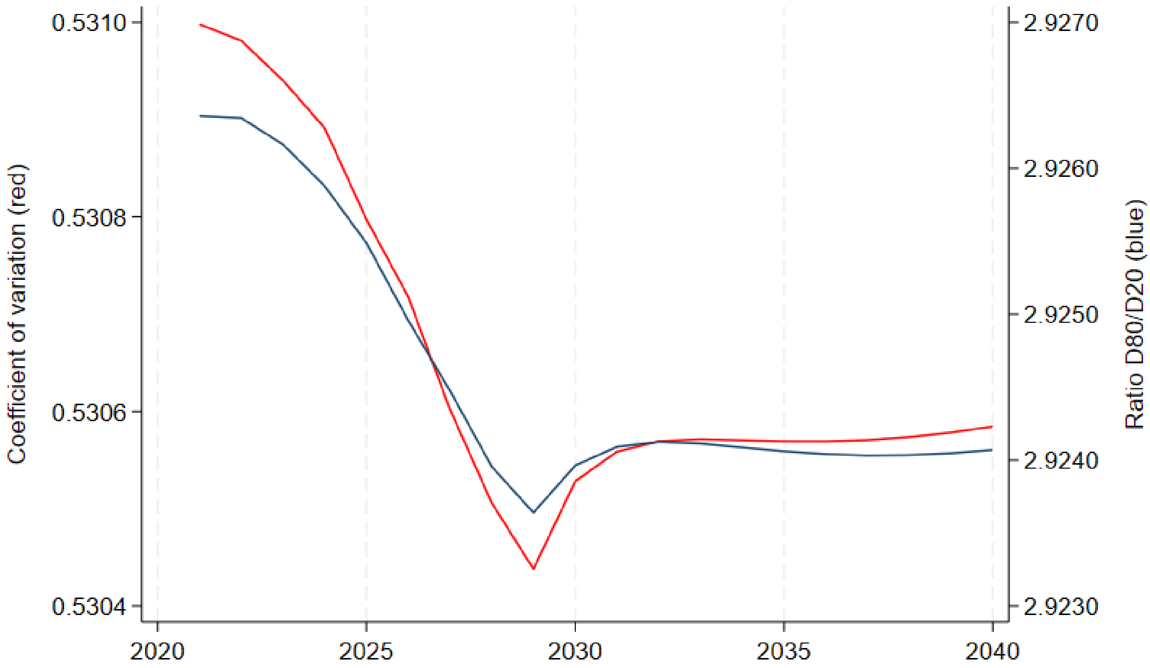
(a) 2029



(a) 2040

Source: RHOMOLO simulations.

Figure C.3. Coefficient of Variation and ratio 80/20 of GDP per capita (% deviations from baseline)



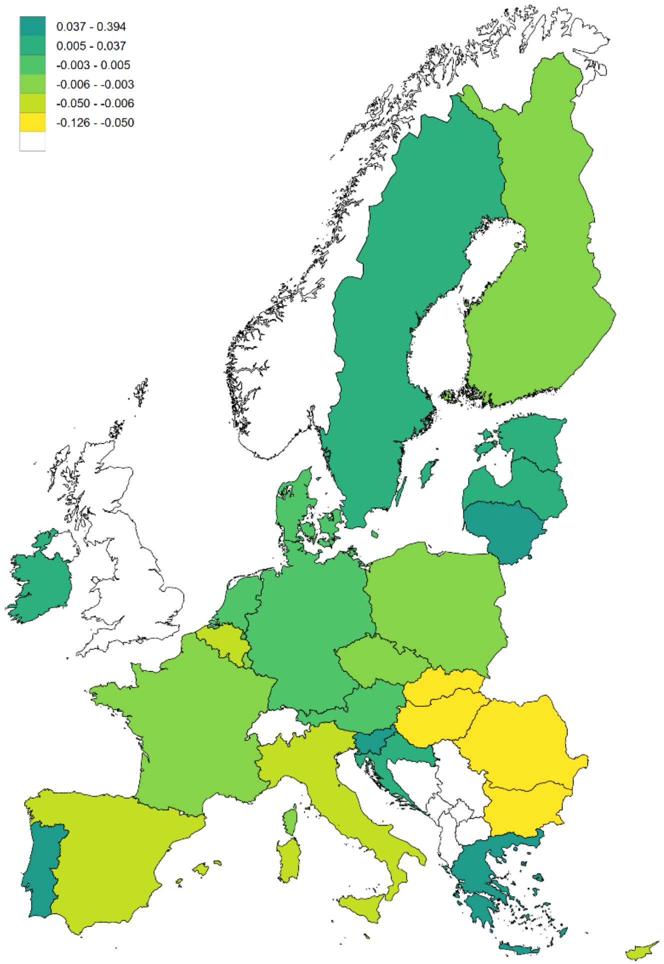
Source: RHOMOLO simulations.

Table C.2. Change in the income distribution in 2040 (% deviations from baseline)

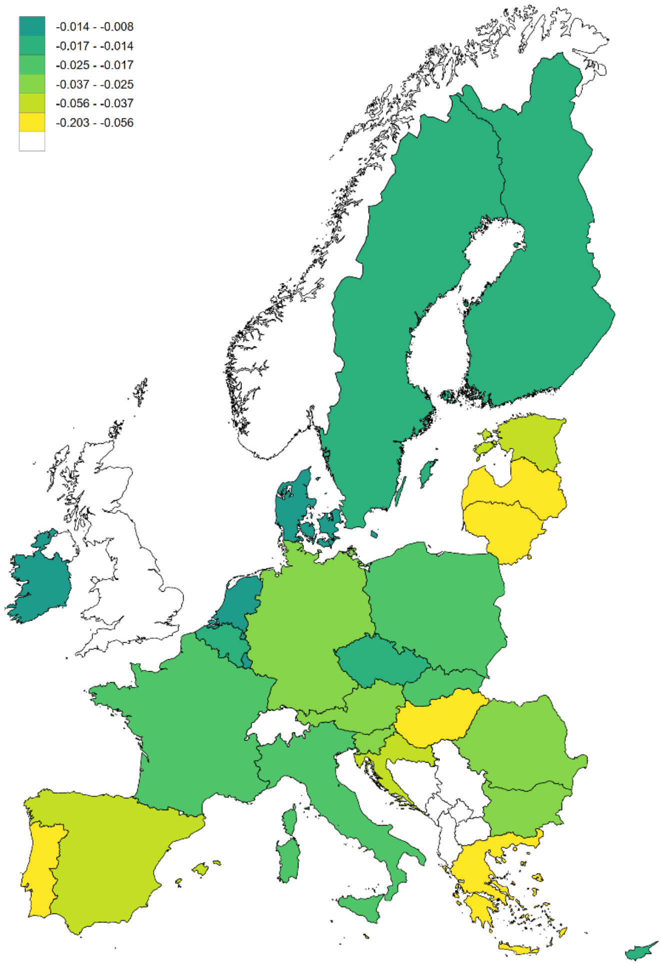
Member State	Bottom 10%	Median (50%)	Top 10%
AT	0.011%	-0.003%	-0.003%
BE	0.008%	0.001%	-0.008%
BG	-0.001%	0.017%	-0.035%
CZ	0.008%	0.003%	-0.031%
DE	0.005%	0.001%	-0.001%
DK	0.005%	0.00%	-0.005%
EL	0.019%	-0.034%	-0.053%
ES	0.012%	0.001%	-0.022%
FI	0.003%	0.004%	-0.006%
FR	0.006%	0.002%	-0.001%
HR	0.006%	0.00%	-0.006%
HU	0.092%	0.065%	-0.272%
IE	0.00%	0.008%	-0.009%
IT	0.02%	-0.005%	-0.017%
LT	0.029%	0.00%	-0.029%
NL	0.002%	0.00%	-0.002%
PL	0.01%	-0.021%	0.007%
PT	0.023%	-0.039%	0.01%
RO	0.017%	0.009%	-0.052%
SE	0.005%	0.004%	-0.01%
SI	0.008%	0.00%	-0.008%
SK	0.024%	0.011%	-0.045%

Source: RHOMOLO simulations.

Figure C.4. Change in the education mismatch (% deviations from baseline)



(a) 2029

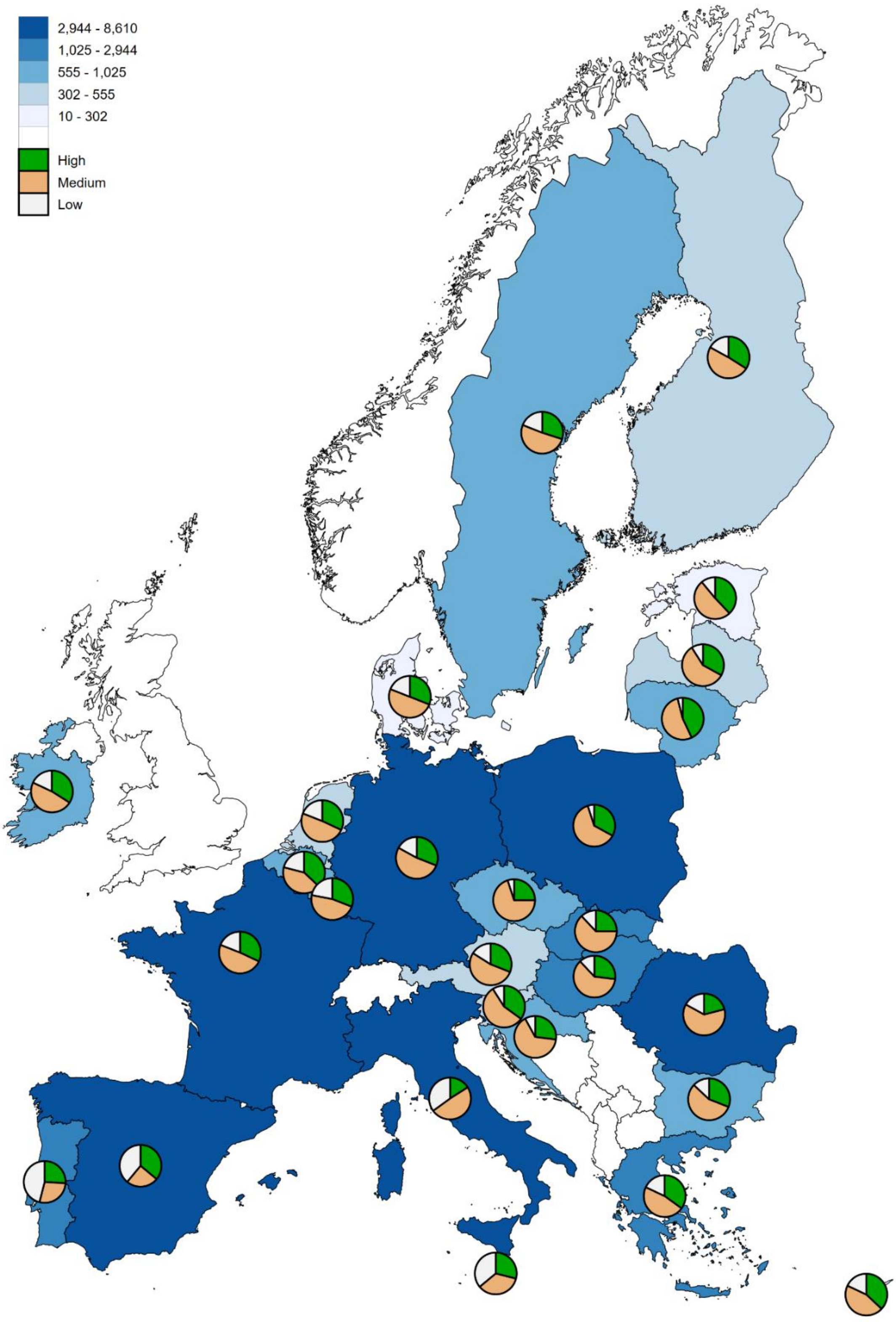


(a) 2040

Source: RHOMOLO simulations.

Annex D. Results for the social inclusion broad area

Figure D.1. ESF+ distribution of funds across Member States and types of labour (€ millions)



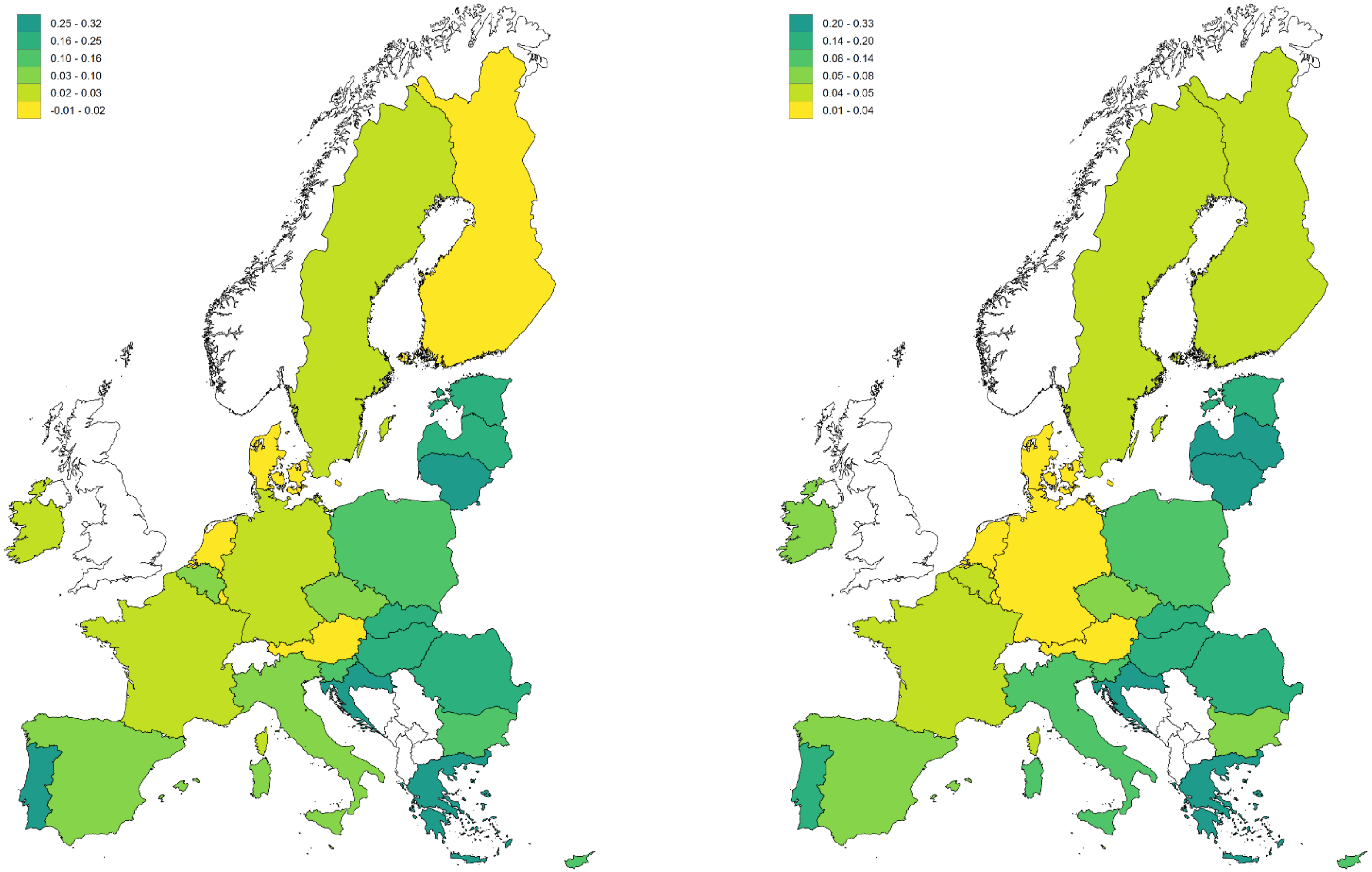
Source: Directorate-General for Employment, Social Affairs and Inclusion.

Table D.1. Planned Allocation of ESF+ funds across Member States, by level of education, thousands

	Education Level				
Member State	**Low (€ 1000)**	**Medium (€ 1000)**	**High (€ 1000)**	**Total**	% of EU total
AT	48	160	94	302	0.66%
BE	208	411	369	988	2.17%
BG	93	424	238	755	1.66%
CY	25	63	52	139	0.31%
CZ	53	718	254	1025	2.26%
DE	814	2431	1430	4675	10.28%
DK	13	33	21	67	0.15%
EE	25	124	93	243	0.53%
EL	355	911	692	1959	4.31%
ES	2112	1376	1934	5423	11.93%
FI	55	153	106	314	0.69%
FR	766	1899	1231	3895	8.57%
HR	64	466	192	723	1.59%
HU	181	936	418	1534	3.37%
IE	100	266	190	555	1.22%
IT	3060	4187	1363	8610	18.93%
LT	28	331	267	626	1.38%
LU	2	5	3	10	0.02%
LV	32	199	113	345	0.76%
MT	21	21	17	58	0.13%
NL	68	172	110	350	0.77%
PL	245	3024	1636	4906	10.79%
PT	1315	815	730	2860	6.29%
RO	526	1813	604	2944	6.47%
SE	154	409	243	806	1.77%
SI	30	180	112	322	0.71%
SK	125	655	260	1041	2.29%
Total	10520	22181	12774	45474	100.00%

Source: Directorate-General for Employment, Social Affairs and Inclusion and authors' calculations.

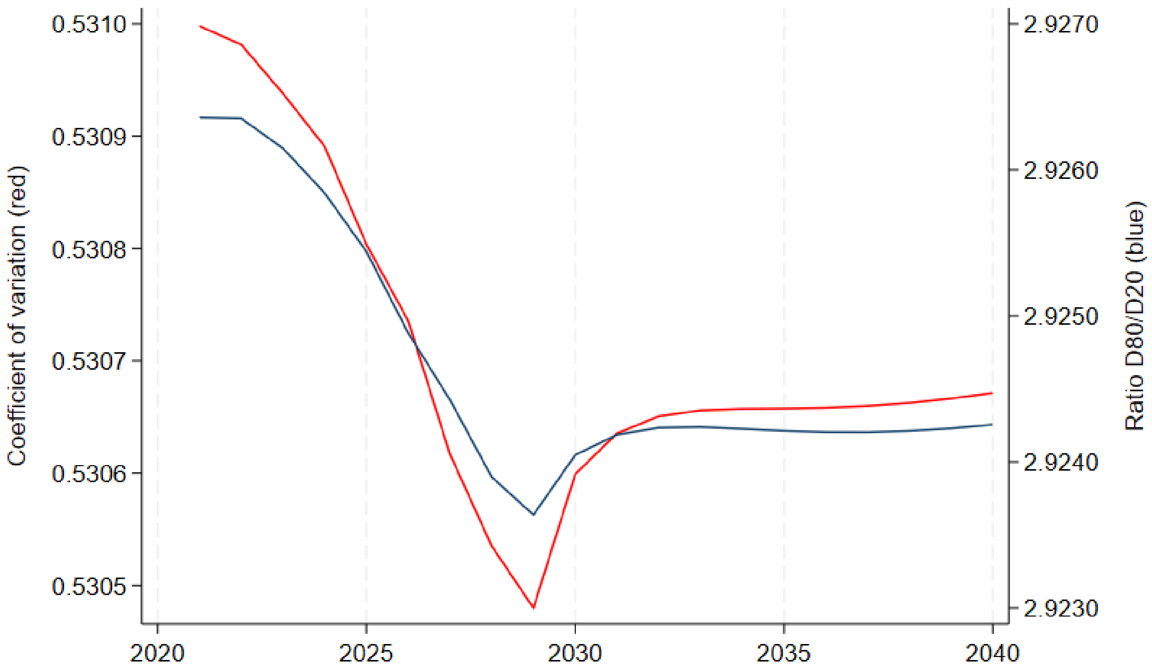
Figure D.2. GDP Impact (% deviations from baseline)



(a) 2029 *(a) 2040*

Source: RHOMOLO simulations.

Figure D.3. Coefficient of Variation and ratio 80/20 of GDP per capita (% deviations from baseline)



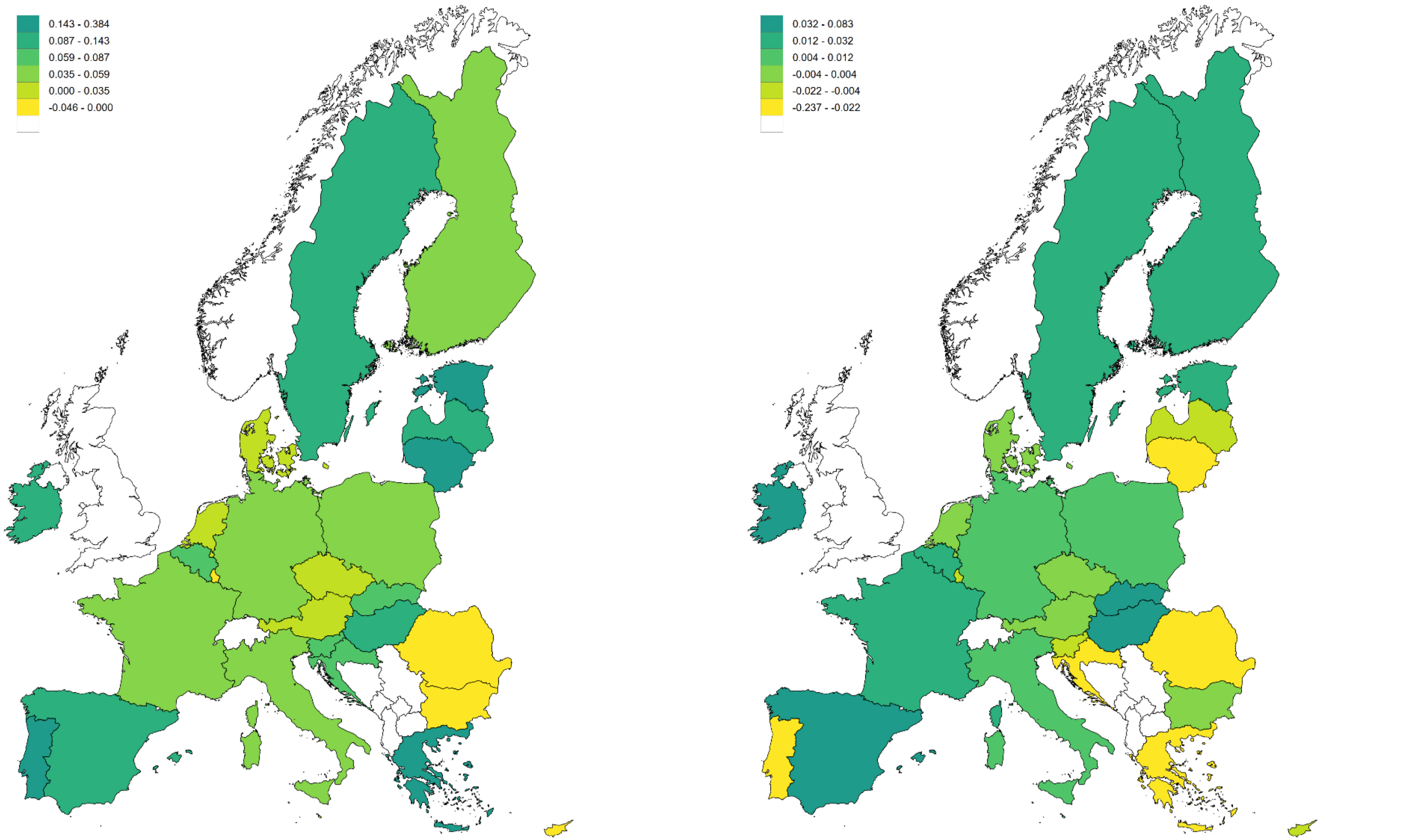
Source: RHOMOLO simulations.

Table D.2. Change in the income distribution in 2040 (% deviations from baseline)

Member State	Bottom 10%	Median (50%)	Top 10%
AT	0.006%	-0.003%	-0.001%
BE	0.009%	0.00%	-0.008%
BG	0.001%	0.011%	-0.025%
CZ	0.007%	0.003%	-0.029%
DE	0.005%	0.001%	0.00%
DK	0.004%	0.00%	-0.004%
EL	0.022%	-0.036%	-0.05%
ES	0.011%	0.003%	-0.03%
FI	0.002%	0.004%	-0.005%
FR	0.005%	0.002%	0.00%
HR	0.005%	0.00%	-0.005%
HU	0.048%	0.037%	-0.145%
IE	0.004%	0.008%	-0.013%
IT	0.022%	-0.005%	-0.018%
LT	0.049%	0.00%	-0.049%
NL	0.002%	0.00%	-0.002%
PL	0.01%	-0.021%	0.007%
PT	0.015%	-0.03%	0.009%
RO	0.02%	0.01%	-0.06%
SE	0.009%	0.004%	-0.015%
SI	0.008%	0.00%	-0.008%
SK	0.046%	0.02%	-0.086%

Source: RHOMOLO simulations.

Figure D.4. Change in the education mismatch (% deviations from baseline)



(a) 2029 *(a) 2040*

Source: RHOMOLO simulations.

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